GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - INGENIUM I4 2.0L PETROL, DTC: POWERTRAIN CONTROL MODULE P0611-02 TO U2300-64 [G2154662]

DESCRIPTION AND OPERATION

POWERTRAIN CONTROL MODULE [PCM] - INGENIUM I4 2.0L PETROL - P0611-02 TO U2300-64

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the Jaguar Land Rover approved diagnostic equipment).
- When performing voltage or resistance tests, always use a digital multimeter that has the resolution ability to view 3 decimal places. For example, on the 2 volts range can measure 1 mV or 2 kΩ range can measure 1 Ω. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Code(s) are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check the JLR claims submission system for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all Diagnostic Trouble Code(s) that could be set in the Powertrain Control Module (PCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: Electronic Engine Controls (303-14C Electronic Engine Controls - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0611-02	Fuel Injector Control Module Performance - General signal failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_P_INJVH3 - ▪ Circuit reference - O_P_INJVL3 - ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P0611-0A	Fuel Injector Control Module Performance - System voltage out of range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_P_INJVH1 - ▪ Circuit reference - O_P_INJVL1 - ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P0611-29	Fuel Injector Control Module Performance - Signal invalid	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_P_INJVH4 - ▪ Circuit reference - O_P_INJVL4 - ▪ Prioritized List of Possible Causes ▪ Other fuel injector related Diagnostic Trouble Code(s) ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0611-64	Fuel Injector Control Module Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_INJVH2 -
		  ▪ Circuit reference - O_P_INJVL2 -
		▪ Prioritized List of Possible Causes
		  ▪ Other fuel injector related Diagnostic Trouble Code(s)
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for fuel injection related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P0615-04	Starter Relay "A" Circuit - System internal failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Internal PCM control fault
		▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P0615-13	Starter Relay "A" Circuit - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_STRL1 -
		▪ Monitor Description
		  ▪ Checking the voltage of the relay control wires to determine a circuit fault
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor relay control circuit open circuit
		  ▪ Starter motor relay disconnected/missing
		  ▪ Starter motor relay failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine start attempt
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check starter motor relay control circuit for open circuit
			  ▪ Check and install a new starter motor relay as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0615-4B	Starter Relay "A" Circuit - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Starting system low side power stage over temperature error (PCM internal hardware error in the driver circuit)
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module detected an internal temperature above the expected range
		  ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Starter motor relay failure
		  ▪ Powertrain control module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Continuous monitor
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new starter motor relay as required
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module
			  ▪ Check core temperature of powertrain control module
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P0615-72	Starter Relay "A" Circuit - Actuator stuck open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ If the starter motor relay is switched on and there is no engine rotation, the PCM registers this Diagnostic Trouble Code
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor fuse has blown
		  ▪ Starter motor relay is stuck open
		  ▪ Starter motor relay control circuit failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine start attempt
			▪ Prioritized Checks to Perform
			  ▪ Check the starter motor fuse
			  ▪ Refer to the electrical circuit diagrams and check starter motor relay control circuit for open circuit, short circuit to ground, short circuit to power, high resistance
			  ▪ Check and install a new starter motor relay as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0615-73	Starter Relay "A" Circuit - Actuator stuck closed	▪ Prioritized List of Possible Causes ▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Monitor runs when at least one starter power stages has turned off ▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new starter motor relay as required ▪ Clear the DTC and retest
P0615-84	Starter Relay "A" Circuit - Signal below allowable range	▪ Prioritized List of Possible Causes ▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Continuous monitor ▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new starter motor relay as required ▪ Clear the DTC and retest
P0615-85	Starter Relay "A" Circuit - Signal above allowable range	▪ Prioritized List of Possible Causes ▪ Starter motor relay failure ▪ Starter motor relay circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Continuous monitor ▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Refer to the electrical circuit diagrams and check starter motor relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check and install a new starter motor relay as required ▪ Clear the DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0616-00	Starter Relay "A" Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_STRH -
		▪ Monitor Description
		  ▪ Starting system high side power stage short circuit to ground
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor relay circuit short circuit to ground, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Monitor runs when ignition switched ON
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the starter motor relay circuit for short circuit to ground, high resistance

P0616-11	Starter Relay "A" Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_STRL1 -
		▪ Monitor Description
		  ▪ Starting system low side power stage short circuit to ground
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor relay circuit short circuit to ground, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Monitor runs when ignition switched ON
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the starter motor relay circuit for short circuit to ground, high resistance

P0617-00	Starter Relay "A" Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_STRH -
		▪ Monitor Description
		  ▪ Starting system high side power stage short circuit to battery
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor relay circuit short circuit to power
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Monitor runs when ignition switched ON
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the starter motor relay circuit for short circuit to power

P0617-12	Starter Relay "A" Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_STRL1 -
		▪ Monitor Description
		  ▪ Starting system low side power stage short circuit to battery
		▪ Prioritized List of Possible Causes
		  ▪ Starter motor relay circuit short circuit to power
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Monitor runs when ignition switched ON
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the starter motor relay circuit for short circuit to power

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P061A-00	Internal Control Module Torque Performance - No sub type information	▪ Prioritized List of Possible Causes ▪ Delivered torque is calculated as being above requested torque within monitoring system. Powertrain control module software error ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Requested torque is always derived from the minimum of the control system request and the corresponding calculation performed within the level 1 monitoring system. Ordinarily under positive torque requests from the driver the monitoring system should always lie above the control system request and hence not perform any limitation. If a software error occurs within the control system the monitoring system shall limit requested torque, if this limitation is active for a significant period of time (that can only be derived from a software error) a fault is judged ▪ Prioritized List of Possible Causes ▪ Powertrain control module failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine speed greater than1200 rpm, fault will only set if limitation is active continuously for 3 minutes to set DTC ▪ Prioritized Checks to Perform ▪ Check and install a new powertrain control module only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P061C-00	Internal Control Module Engine RPM Performance - No sub type information	▪ Prioritized List of Possible Causes ▪ Desired ignition angle not plausible. Powertrain control module software error ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P061D-42	Internal Control Module Engine Air Mass Performance - General memory failure	▪ Prioritized List of Possible Causes
		  ▪ Injector cut off patterns found not plausible. Powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P061D-43	Internal Control Module Engine Air Mass Performance - Special memory failure	▪ Prioritized List of Possible Causes
		  ▪ Desired fueling control limits implausible. Powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P061D-62	Internal Control Module Engine Air Mass Performance - Signal compare failure	▪ Prioritized List of Possible Causes
		  ▪ Predicted air mass was found not plausible. Powertrain control module software error
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P061F-00	Internal Control Module Throttle Actuator Controller Performance - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Not applicable
		▪ Monitor Description
		  ▪ Measured throttle position deviates from requested position excessively
		▪ Prioritized List of Possible Causes
		  ▪ Other throttle related Diagnostic Trouble Code(s)
		  ▪ Blockages/restriction preventing free motion of the throttle blade
		  ▪ Throttle actuator failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition off - On and wait 60seconds prior to starting engine, limp home position check will be performed if blade does not achieve a position close to the learnt limp home position a fault is set as the throttle is faulty
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first
			  ▪ Inspect throttle for signs of blockage or restriction that prevent free motion of the throttle blade
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P0627-07	Fuel Pump "A" Control Circuit/Open - Mechanical failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ To detect a fault on low pressure pump
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Harness failure between fuel pump driver module and low pressure fuel pump
		  ▪ Low pressure fuel pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Low pressure fuel pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the low pressure fuel pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new low pressure fuel pump as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P062A-31	Fuel Pump "A" Control Circuit Range/Performance - No signal	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ To detect a fault on low pressure pump
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel pump driver module signal circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel pump driver module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel pump driver module signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new fuel pump driver module as required

P062F-00	Internal Control Module EEPROM Error - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ The fault is set when at least three software blocks cannot be read
		▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest. Check if DTC is re-set
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P062F-43	Internal Control Module EEPROM Error - Special memory failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ The fault is set when one software block can not be written more than 3 times
		▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest. Check if DTC is re-set
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0633-00	Immobilizer Key Not Programed - ECM/PCM - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Signals an error reading or writing the Target ID (TID) to the PCM EEPROM)
		▪ Prioritized List of Possible Causes
		  ▪ Power is lost from the powertrain control module or the body control module during the immobilizer learn routine
		  ▪ Security key invalid
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the powertrain control module and body control module
			  ▪ Re-configure the powertrain control module using the Jaguar Land Rover approved diagnostic equipment. Re-configure the instrument cluster using the Jaguar Land Rover approved diagnostic equipment
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the immobilization procedure
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest. Check if DTC is re-set
			  ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P0634-00	Control Module Internal Temperature "A" Too High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ECM -
		▪ Monitor Description
		  ▪ A defect is detected when internal powertrain control module temperature is greater than 119°C
		▪ Prioritized List of Possible Causes
		  ▪ Ventilation of the ambient air surrounding powertrain control module cooling fan
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module cooling fan fuse failure
		  ▪ Powertrain control module cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Powertrain control module cooling fan failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Check for blockage and good ventilation around powertrain control module cooling fan
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the powertrain control module cooling fan fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new powertrain control module cooling fan only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P064A-42	Fuel Pump Control Module "A" - General memory failure	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pump driver module diagnostic line circuit open circuit, short circuit to power, short circuit to ground ▪ Fuel pump driver module failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the fuel pump driver module diagnostic circuit for open circuit, short circuit to power, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest ▪ Check and install a new fuel pump driver module as required
P064A-98	Fuel Pump Control Module "A" - Component or system over temperature	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Fuel pump driver module diagnostic line circuit open circuit, short circuit to power, short circuit to ground ▪ Fuel pump driver module failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the fuel pump driver module diagnostic circuit for open circuit, short circuit to power, short circuit to ground ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest ▪ Check and install a new fuel pump driver module as required
P064D-01	Internal Control Module O2 Sensor Processor Performance Bank 1 - General electrical failure	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P064D-02	Internal Control Module O2 Sensor Processor Performance Bank 1 - General signal failure	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0658-00	Actuator Supply Voltage "A" Circuit Low - No sub type information	Prioritized List of Possible Causes - Other related Diagnostic Trouble Code(s) - Harness fault - powertrain control module power or ground supply - Vehicle battery power or ground distribution fault - Vehicle battery fault - Charging system fault - under charging	Prioritized Checks to Perform - Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index - Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests - Check the vehicle charging system performance to make sure the voltage regulation is correct - Refer to the electrical circuit diagrams and check the powertrain control module power and ground circuits - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P0659-00	Actuator Supply Voltage "A" Circuit High - No sub type information	Prioritized List of Possible Causes - Other related Diagnostic Trouble Code(s) - Vehicle battery over charged - Charging system fault - over charging - Generator circuit short circuit to ground, short circuit to power, open circuit, high resistance - Powertrain control module main fuse mechanical integrity - Connector is disconnected, connector pin is backed out, connector pin corrosion - Harness failure - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance	Prioritized Checks to Perform - Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index - Check the vehicle charging system performance to make sure the voltage regulation is correct - Refer to the electrical circuit diagrams and check the generator circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check powertrain control module main fuse for mechanical integrity - Inspect connectors for signs of water ingress and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0686-11	ECM/PCM Power Relay Control Circuit Low - Circuit short to ground	▪ Control Module Cavity
		  ▪ Circuit reference - O_S_MRLY Control -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
		  ▪ Engine management system relay power and ground circuits open circuit
		  ▪ Engine management system relay circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine management system relay failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check engine management system relay power and ground circuits for open circuit
			  ▪ Refer to the electrical circuit diagrams and check the engine management system relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new engine management system relay as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P0687-24	ECM/PCM Power Relay Control Circuit High - Signal stuck high	▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module power relay circuit short circuit to power, short circuit to another circuit
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module power relay circuit for short circuit to power, short circuit to another circuit

P068A-00	ECM/PCM Power Relay De-Energized Performance - Too Early - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module main power relay de-energized before normal shutdown processes
		  ▪ Power and ground connections to the powertrain control module, open circuit, high resistance
		  ▪ Battery disconnection or relay control circuit intermittent harness failure
		  ▪ Powertrain control module power relay failure
			▪ Prioritized Checks to Perform
			  ▪ This DTC is set when the engine management system high current relay contacts open early - Indicating a power hold failure. Check the vehicle battery has not been disconnected before the engine management system relay has powered down. Check the operation of the engine management system high current relay. Refer to the electrical circuit diagrams and check the engine management system high current relay supply and control circuits for open circuits, high resistance, short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Clear DTC and retest
			  ▪ Check and install a new relay as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P06A6-00	Sensor Reference Voltage "A" Circuit Range/Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_V_5VRAILPS - ▪ Circuit reference - O_5VCLNTP - ▪ Circuit reference - O_V_5VMAP - ▪ Circuit reference - O_V_5VBPS - ▪ Circuit reference - O_V_5VCRS - ▪ Circuit reference - O_V_5VLPEGR - ▪ Circuit reference - O_V_5VAPP1 - ▪ Circuit reference - O_V_5VWGPT - ▪ Circuit reference - O_V_5VAMS - ▪ Prioritized List of Possible Causes ▪ Powertrain control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Prioritized Checks to Perform ▪ Refer to electrical circuit diagrams and check the powertrain control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
P06A7-00	Sensor Reference Voltage "B" Circuit Range/Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_V_5VGPFDP - ▪ Circuit reference - O_V_5VFLPS - ▪ Circuit reference - O_V_5VFTPS - ▪ Circuit reference - O_V_5VCLUPOT - ▪ Circuit reference - O_V_5VAPP2 - ▪ Circuit reference - O_V_5VLPEGRDP - ▪ Prioritized List of Possible Causes ▪ Powertrain control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Prioritized Checks to Perform ▪ Refer to electrical circuit diagrams and check the powertrain control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P06A8-00	Sensor Reference Voltage "C" Circuit Range/Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5PHEV -
		  ▪ Circuit reference - O_V_5VTVA -
		  ▪ Circuit reference - O_V_5VGPS -
		▪ Prioritized List of Possible Causes
		  ▪ Powertrain control module sensor 5 volt supply circuit short circuit to ground, short circuit to power, open circuit
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the powertrain control module sensor 5 volt supply circuit for short circuit to ground, short circuit to power, open circuit
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion

P06D1-04	Internal Control Module Ignition Coil Control Performance - System internal failures	▪ Prioritized List of Possible Causes
		  ▪ Other ignition coil related Diagnostic Trouble Code(s)
		  ▪ Powertrain control module power and ground circuits open circuit
		  ▪ Powertrain control module failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module power and ground circuits for open circuit
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check the powertrain control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Check and install a new powertrain control module only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment clear the DTC and retest

P06DA-13	Engine Oil Pressure Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_OILP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Variable oil pump solenoid fuse failure
		  ▪ Variable oil pump solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Variable oil pump solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the variable oil pump solenoid fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the variable oil pump solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new variable oil pump solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P06DA-4B	Engine Oil Pressure Control Circuit/Open - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_OILP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Variable oil pump solenoid fuse failure
		  ▪ Variable oil pump solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Variable oil pump solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the variable oil pump solenoid fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the variable oil pump solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new variable oil pump solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P06DB-11	Engine Oil Pressure Control Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_OILP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Variable oil pump solenoid fuse failure
		  ▪ Variable oil pump solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Variable oil pump solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the variable oil pump solenoid fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the variable oil pump solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new variable oil pump solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P06DC-12	Engine Oil Pressure Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_OILP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Variable oil pump solenoid fuse failure
		  ▪ Variable oil pump solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Variable oil pump solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the variable oil pump solenoid fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the variable oil pump solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new variable oil pump solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P06E9-07	Engine Starter Performance - Mechanical failures	- Purpose of the DTC
		  - To indicate a starter solenoid circuit fault
		- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Not applicable
		- Prioritized List of Possible Causes
		  - Car configuration file mismatch with vehicle specification
		  - Starter solenoid pinion circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Starter solenoid failure
			- Prioritized Checks to Perform
			  - Using the Jaguar Land Rover approved diagnostic equipment, run application - Service / Select CCF / Copy from AS BUILT
			  - Refer to the electrical circuit diagrams and check the starter solenoid pinion circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Test the starter motor for correct operation. Follow all the instructions in the starter motor testing procedure. For additional information, refer to: **Starter Motor Testing** (303-06C Starting System - INGENIUM I4 2.0L Petrol, General Procedures). Only replace the starter motor when it is diagnosed as being faulty or inoperative
			  - Install a new starter solenoid

P0703-62	Brake Switch "B" Circuit - Signal compare failure	- Prioritized List of Possible Causes
		  - The powertrain control module detected failure when comparing two or more input parameters for plausibility
		  - The brake pedal switch signal received over CAN is inoperative
		  - Brake pedal switch detached from pedal box while electrically connected
		  - Brake pedal switch adjustment incorrect
		  - Brake pedal switch harness failure
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Brake pedal switch failure
			- Prioritized Checks to Perform
			  - Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test
			  - Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  - Refer to the electrical circuit diagrams and check the brake pedal switch circuits for open circuit, short circuit to ground, short circuit to power
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Check and install a new brake pedal switch as required
			  - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P0850-00	Park/Neutral Switch Input Circuit - No sub type information	- Prioritized List of Possible Causes
		  - Hardwired park/neutral switch does not match the information within the CAN signal
		  - Park/neutral switch circuit, short circuit to power, short circuit to ground, open circuit
		  - HS CAN network failure
			- Prioritized Checks to Perform
			  - Refer to the electrical circuit diagrams and check the park/neutral switch circuit, for short circuit to power, short circuit to ground, open circuit
			  - If this DTC is set with other HS CAN related Diagnostic Trouble Code(s) refer to the electrical circuit diagrams and check HS CAN network for short circuit to power, short circuit to ground, open circuit

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P0A0F-07	Engine Failed To Start - Mechanical failures	▪ Prioritized List of Possible Causes ▪ Car configuration file incorrect for stop/start system ▪ Harness failure - Wiring integrity starter motor circuit ▪ Mechanical starter motor failure ▪ Starter motor pinion not operating ▪ Harness failure - Wiring integrity camshaft sensor circuit ▪ Harness failure - Wiring integrity crankshaft sensor circuit	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required ▪ Test the starter motor for correct operation. Follow all the instructions in the starter motor testing procedure. For additional information, refer to: **Starter Motor Testing** (303-06C Starting System - INGENIUM I4 2.0L Petrol, General Procedures). Only replace the starter motor when it is diagnosed as being faulty or inoperative ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Clear the DTC and retest
P0A14-13	Engine Mount "A" Control Circuit/Open - Circuit open	▪ Control Module Cavity ▪ Circuit reference - O_T_EMC - ▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined an open circuit through lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Engine mount - Active - left/right circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine mount - Active - left/right failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the engine mount - Active - left/right circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new engine mount - Active - left/right as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P0A15-11	Engine Mount "A" Control Circuit Low - Circuit short to ground	▪ Control Module Cavity ▪ Circuit reference - O_T_EMC - ▪ Prioritized List of Possible Causes ▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Engine mount - Active - left/right circuit short circuit to ground, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine mount - Active - left/right failure	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the engine mount - Active - left/right circuit for short circuit to ground, high resistance ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new engine mount - Active - left/right as required ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P0A16-12	Engine Mount "A" Control Circuit High - Circuit short to battery	▪ Control Module Cavity
		  ▪ Circuit reference - O_T_EMC -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
		  ▪ Engine mount - Active - left/right circuit short circuit to power
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Engine mount - Active - left/right failure
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the engine mount - Active - left/right circuit for short circuit to power
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new engine mount - Active - left/right as required
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P115F-11	Electronic Control Module Cooling Fan Circuit - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ECM -
		▪ Monitor Description
		  ▪ Monitor to diagnose short circuit to ground
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module cooling fan circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the powertrain control module cooling fan fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary

P115F-12	Electronic Control Module Cooling Fan Circuit - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_ECM -
		▪ Monitor Description
		  ▪ Monitor to diagnose short circuit to power
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module cooling fan fuse failure
		  ▪ Powertrain control module cooling fan circuit short circuit to power, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the powertrain control module cooling fan fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for short circuit to power, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P115F-13	Electronic Control Module Cooling Fan Circuit - Circuit open	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_ECM - ▪ Monitor Description ▪ Monitor to diagnose open circuit ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module cooling fan fuse failure ▪ Powertrain control module cooling fan circuit open circuit, high resistance	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, start engine and idle for 30 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check the integrity of the powertrain control module cooling fan fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the powertrain control module cooling fan circuit for open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P1315-00	Persistent Misfire - No sub type information	**NOTE:**
		The misfire monitor runs continuously and is designed to detect levels of misfire that can cause either thermal damage to the catalyst or excessive tailpipe emissions. Determination of a misfire is made by analysis of changes in crankshaft speed, since a misfire will cause a fall in speed. Essentially the monitor determines the time taken for the crank shaft to move through 180 degrees (4 cylinder only) for the current firing event and compares that to the time taken for the previous firing event. If the time taken for the 'current' firing is greater than the previous firing, then a misfire is judged to have occurred
		
		- Any of the following Diagnostic Trouble Code(s) if set will cause a misfire. These must be repaired before additional investigations are conducted
		  - P0030-00 P0031-00 P0032-00 P00D1-00 P0131-00 P0132-00 P0133-00 P0134-00 P0192-00 P0193-00 P0444-00 P0458-00 P0459-00 P0496-00 P0497-00 P064D-08 P064D-17 P064D-16 P064D-86 P2096-00 P2096-02 P2097-00 P2097-02 P2231-00 P2237-00 P2237-02 P2237-64 P2243-00 P2251-00 P228C-77 P228D-77 P2626-00
		- The above Diagnostic Trouble Code(s) are relating to the following systems or components
		  - Ignition system
		  - Fuel system
		  - Charge air system
		  - Evaporative emissions canister purge valve
		- Prioritized List of Possible Causes
		  - Fuel level low
		  - Spark plug(s) fouled or failed
		  - Reluctor ring to sensor runout and air gap out of tolerance
		  - Crankshaft position sensor incorrectly installed
		  - Poor fuel quality
		  - Incorrect fuel grade or blend
		  - Contaminated fuel or unauthorized additives
		  - Driveline imbalance - Debris around rotating components, driveline components excessive wear, wheel and tire imbalance
		  - Harness failure - Wiring integrity
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			
			- **WARNING:**
			When completing the cylinder compression tests, if the test result is 0%, or no pressure, **DO NOT** immediately replace the engine
			
			NOTE:
			The 0% reading on the compression test could be due to the inlet valves not opening, caused by a mechanical failure in the CVVL unit
			
			When instructed to complete the cylinder compression test for this DTC, refer to the warning and note above this text
			- Check powertrain control module for any of the shown Diagnostic Trouble Code(s) and relating to the following systems or components. These must be repaired before additional investigations are conducted
			  - Ignition system
			  - Fuel system
			  - Charge air system
			  - Evaporative emissions canister purge valve
			- Prioritized Checks to Perform
			  - Make sure vehicle has sufficient fuel
			  - Check spark plug(s) for fouling or failure, install new as required
			  - Check reluctor ring for correct installation
			  - Check crankshaft position sensor for correct installation
			  - Drain and refuel with new fuel of correct grade and known quality
			  - Check driveline for debris around rotating components
			  - Check driveline for excessive wear or backlash
			  - Check wheels and tires for snow or mud packing and correct balance
			  - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
			  - Continuous Variable Vale Lift (CVVL) system - mechanical failure. If the vehicle has the CVVL system fitted to the engine, remove the CVVL unit and inspect the components for mechanical damage or a broken spring. Replace as necessary
			  - Check cylinder compressions both cold and hot

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
		▪ Engine misfire caused by a Continuous Variable Valve Lift (CVVL) mechanical failure ▪ Low cylinder compressions - Worn damaged engine components, worn piston rings, incorrect valve clearance, worn valves ▪ Catalyst blocked or damaged ▪ Engine mount damaged ▪ Transmission mount damaged ▪ Accessory drive belt components incorrectly installed ▪ Fuel injector contamination or blocked ▪ Water or condensation in charge air cooler or charge air cooler outlet pipe (pre throttle) ▪ Engine build excessive wear	▪ Check for correct valve clearance ▪ Check catalyst for damage or blockage ▪ Check engine mounts for damage ▪ Check transmission mounts for damage ▪ Check accessory drive belt components are correctly installed ▪ Check fuel injector for contamination or blockage ▪ Check charge air cooler and charge air cooler outlet pipe for water or condensation, drain as required ▪ Check engine oil pressure is to specification ▪ For further detailed information, please refer to the relevant section of the workshop manual. Section 303 - Electronic Engine Controls
P131A-00	Low Fuel Level Detection - No sub type information	• Prioritized List of Possible Causes ▪ Low level fuel condition ▪ Other fuel related Diagnostic Trouble Code(s) are set	• Prioritized Checks to Perform ▪ Check the fuel level, add fuel if required and clear the DTC ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
P1338-38	Fuel Pump Driver Module Communication Circuit (Fuel Pump Driver Module) - Signal frequency incorrect	• Prioritized List of Possible Causes ▪ The fuel pump driver module has observed more than ±15% deviation in the PWM frequency from the powertrain control module for more than 6 seconds ▪ Signal frequency should be greater than 127.5 hz and less than 172.5 hz. Voltage should between 0 - 12 V ▪ Fuel pump driver module PWM control circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel pump driver module fault ▪ Powertrain control module fault	**NOTE:** This fault does not indicate an issue with the fuel driver module. The fuel driver module must not be replaced. • Prioritized Checks to Perform ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests ▪ Refer to electrical circuit diagrams and check the fuel pump driver module PWM control circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved Picoscope, monitor the PWM signal at the fuel pump driver module, with the engine at idle. The signal frequency should be greater than 127.5 hz and less than 172.5 hz. Voltage should be square wave with 0 - battery voltage amplitude ▪ If the observed frequency or voltage values are outside these tolerances, replace the fuel pump driver module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P144B-00	Evaporative Emission System Secondary Purge Vapor Line Restricted/Blocked - No sub type information	▪ Prioritized List of Possible Causes ▪ Purge control valve circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Pipework venturi to upper air duct blocked ▪ Pipework purge control valve to venturi blocked/disconnected ▪ Pipework pre throttle to venturi blocked	▪ Prioritized Checks to Perform ▪ Refer to electrical circuit diagrams and check the purge control valve circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check evaporative emission system purge control valve pipework for blocked or disconnected
P144C-00	Evaporative Emission System Purge Flow Performance During Boost - No sub type information	▪ Prioritized List of Possible Causes ▪ Second purge control valve pipe disconnected ▪ Fuel system leakage ▪ Other fuel system related issue, causing change in fuel adaptions	**NOTE:** This DTC may set as a result of P0191-85 setting. Complete the repair actions for P0191-85 first, clear the Diagnostic Trouble Code(s) and retest. If this DTC returns, perform the checks below ▪ Prioritized Checks to Perform ▪ Check second purge control valve pipe is not disconnected ▪ Check powertrain control module for fuel related Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete smoke test on purge system. Confirm no leakage. Repair if necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete smoke test on intake air system - Air box side. Confirm no leakage. Repair if necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the routine 0x0407 - Fuel system prime. Check for any leaks in fuel system. Repair if necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the routine 0x0406 - Powertrain control module adaption clear. This routine will reset all adaptions including the fuel adaptions diagnostic ▪ Complete a standard road test to confirm the issue is repaired

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P162F-00	Starter Motor Disabled - Engine Crank Time Too Long - No sub type information	▪ Prioritized List of Possible Causes ▪ Other starting related Diagnostic Trouble Code(s) ▪ Vehicle battery failure	▪ Prioritized Checks to Perform ▪ Check powertrain control module for starting related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and make sure it is fully charged and serviceable before performing further tests ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P167F-00	Non-OEM Calibration Detected - No sub type information	▪ Prioritized List of Possible Causes ▪ This DTC will indicate that the powertrain control module software was modified and the parameters are or were no longer the designated software parameters designed for that particular engine ▪ Mismatch between car configuration file and powertrain control module calibration	**NOTE:** This DTC cannot be erased. It is a permanent DTC that has been set in the EPROM for traceability only ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and install latest relevant level of software to the powertrain control module ▪ If further assistance is required to resolve this DTC submit a Technical Assistance request
P168F-00	Cold Start Injection System Performance - No sub type information	▪ Prioritized List of Possible Causes ▪ Other related Diagnostic Trouble Code(s) ▪ Fuel injector failure ▪ Powertrain control module failure	▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Refer to the electrical circuit diagrams and check fuel injector connections are secure and wiring integrity ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest ▪ Allow an overnight soak and complete a cold start. Check powertrain control module for this DTC. If the DTC has reset post the cold start, check and install a new powertrain control module as required. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2088-00	"A" Camshaft Position Actuator Control Circuit Low Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_VVTIN - ▪ Monitor Description ▪ Inlet camshaft actuator short to ground ▪ Prioritized List of Possible Causes ▪ Inlet camshaft solenoid circuit fault ▪ Inlet camshaft solenoid internal failure	▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy ▪ Start and idle engine for at least 2 seconds ▪ Prioritized Checks to Perform ▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest ▪ If the same DTC persists (P2088-00), then check the connector and wiring between the PCM and the inlet camshaft solenoid for short circuit to ground. Repair wiring as required ▪ If an exhaust camshaft solenoid DTC is reported (P2090-00), then replace the inlet camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair
P2089-00	"A" Camshaft Position Actuator Control Circuit High Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_VVTIN - ▪ Monitor Description ▪ Inlet camshaft actuator short to battery ▪ Prioritized List of Possible Causes ▪ Inlet camshaft solenoid circuit fault ▪ Inlet camshaft solenoid internal failure	▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy ▪ Start and idle engine for at least 2 seconds ▪ Prioritized Checks to Perform ▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest ▪ If the same DTC persists (P2089-00), then check the connector and wiring between the PCM and the inlet camshaft solenoid for short circuit to power. Repair wiring as required ▪ If an exhaust camshaft solenoid DTC is reported (P2091-00), then replace the inlet camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair
P2090-00	"B" Camshaft Position Actuator Control Circuit Low Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_VVTEX - ▪ Monitor Description ▪ Exhaust camshaft actuator bank 1 short to ground ▪ Prioritized List of Possible Causes ▪ Exhaust camshaft solenoid circuit fault ▪ Exhaust camshaft solenoid internal failure	▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy ▪ Start and idle engine for at least 2 seconds ▪ Prioritized Checks to Perform ▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest ▪ If the same DTC persists (P2090-00), then check the connector and wiring between the PCM and the exhaust camshaft solenoid for short circuit to ground. Repair wiring as required ▪ If an inlet camshaft solenoid DTC is reported (P2088-00), then replace the exhaust camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2091-00	"B" Camshaft Position Actuator Control Circuit High Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_VVTEX -
		▪ Monitor Description
		  ▪ Exhaust camshaft actuator bank 1 short to power
		▪ Prioritized List of Possible Causes
		  ▪ Exhaust camshaft solenoid circuit fault
		  ▪ Exhaust camshaft solenoid internal failure
			▪ Vehicle Conditions to Enable Diagnostic Trouble Code Setting Strategy
			  ▪ Start and idle engine for at least 2 seconds
			▪ Prioritized Checks to Perform
			▪ Clear the DTC. Refer to the electrical circuit diagrams and swap the camshaft solenoid and the exhaust solenoid circuit connectors then retest
			  ▪ If the same DTC persists (P2091-00), then check the connector and wiring between the PCM and the exhaust camshaft solenoid for short circuit to power. Repair wiring as required
			  ▪ If an inlet camshaft solenoid DTC is reported (P2089-00), then replace the exhaust camshaft solenoid. Reconnect the correct corresponding circuit connectors and retest to confirm the repair

P2096-00	Post Catalyst Fuel Trim System Too Lean Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Detects when secondary fueling adaption offset is too large for fuel mixture control to be maintained
		▪ Prioritized List of Possible Causes
		  ▪ Exhaust system leak
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Mid catalyst oxygen sensor incorrectly installed or harness failure
		  ▪ Pre catalyst oxygen sensor incorrectly installed or harness failure
		  ▪ Mid catalyst oxygen sensor damaged or contaminated internally
		  ▪ Pre catalyst oxygen sensor damaged or contaminated internally
		  ▪ Mid catalyst oxygen sensor failure
		  ▪ Pre catalyst oxygen sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Drive vehicle under steady state conditions for up to 20 minutes
			  ▪ If sensor changed drive vehicle under steady-state conditions to allow vehicle to adapt to new sensor. Fault may still be present on vehicle. Perform fault clear after drive. If fault persists repeat drive and clear fault again
			▪ Prioritized Checks to Perform
			  ▪ Check the exhaust system for leaks. Repair as necessary
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check pre and mid catalyst oxygen sensors for damaged and correct installation
			  ▪ Refer to the electrical circuit diagrams and check the mid catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance
			  ▪ Refer to the electrical circuit diagrams and check the pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new pre catalyst oxygen sensor as required
			  ▪ Check and install a new mid catalyst oxygen sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2097-00	Post Catalyst Fuel Trim System Too Rich Bank 1 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Detects when secondary fueling adaption offset is too small for fuel mixture control to be maintained ▪ Prioritized List of Possible Causes ▪ Exhaust system leak ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Mid catalyst oxygen sensor incorrectly installed or harness failure ▪ Pre catalyst oxygen sensor incorrectly installed or harness failure ▪ Mid catalyst oxygen sensor damaged or contaminated internally ▪ Pre catalyst oxygen sensor damaged or contaminated internally ▪ Mid catalyst oxygen sensor failure ▪ Pre catalyst oxygen sensor failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle under steady state conditions for up to 20 minutes ▪ If sensor changed drive vehicle under steady-state conditions to allow vehicle to adapt to new sensor. Fault may still be present on vehicle. Perform fault clear after drive. If fault persists repeat drive and clear fault again ▪ Prioritized Checks to Perform ▪ Check the exhaust system for leaks. Repair as necessary ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check pre and mid catalyst oxygen sensors for damaged and correct installation ▪ Refer to the electrical circuit diagrams and check the mid catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ▪ Refer to the electrical circuit diagrams and check the pre catalyst oxygen sensor circuits for open circuit short circuit to power, short circuit to ground, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest ▪ Check and install a new pre catalyst oxygen sensor as required ▪ Check and install a new mid catalyst oxygen sensor as required
P2100-00	Throttle Actuator "A" Control Motor Circuit/Open - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_TVMPOS - ▪ Circuit reference - O_T_TVMNEG - ▪ Monitor Description ▪ Throttle internal power stage is deemed open circuit through internal hardware checks which commence when throttle control PWM duty is excessive ▪ Prioritized List of Possible Causes ▪ Obstruction in the electric throttle unit motor assembly ▪ Electric throttle unit circuit, open circuit ▪ Electric throttle unit motor internal failure - open circuit ▪ Powertrain control module internal throttle motor control stage failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Battery voltage greater than 8 volts ▪ No throttle driver error present ▪ Throttle adaptions completed ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle ▪ Prioritized Checks to Perform ▪ Check for obstruction in the electric throttle unit motor assembly ▪ Refer to the electrical circuit diagrams and check the electric throttle unit circuit for open circuit ▪ Check the electric throttle unit motor for open circuit ▪ Repair wiring harness as required. Clear the DTC and retest. If the DTC resets check and install a new powertrain control module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2101-00	Throttle Actuator "A" Control Motor Circuit Range/Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_TVMPOS - ▪ Circuit reference - O_T_TVMNEG - ▪ Monitor Description ▪ Throttle internal power stage is unable to communicate internal fault status to software processor due to internal hardware fault when throttle control PWM duty is excessive ▪ Prioritized List of Possible Causes ▪ Electric throttle unit - Circuit O_T_TVMPOS - and - Circuit O_T_TVMNEG - short circuit together ▪ Electric throttle unit - Circuit O_T_TVMPOS for short circuit to ground ▪ Electric throttle unit - Circuit O_T_TVMNEG - short circuit to power ▪ Electric throttle unit motor internal failure - short circuit together ▪ Powertrain control module internal throttle motor control stage failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Battery voltage greater than 8 volts ▪ No throttle driver error present ▪ Throttle adaptions completed ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the electric throttle unit - Circuit O_T_TVMPOS - and - Circuit O_T_TVMNEG for short circuit together ▪ Refer to the electrical circuit diagrams and check the electric throttle unit - Circuit O_T_TVMPOS for short circuit to ground ▪ Refer to the electrical circuit diagrams and check the electric throttle unit - Circuit O_T_TVMNEG for short circuit to power ▪ Disconnect harness from electric throttle unit and check electric throttle unit motor internal resistance is greater than 100 ohms ▪ Repair wiring harness as required. Clear the DTC and retest. If the DTC resets check and install a new powertrain control module as required
P2103-85	Throttle Actuator "A" Control Motor Circuit High - Signal above allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_T_TVMPOS - ▪ Circuit reference - O_T_TVMNEG - ▪ Monitor Description ▪ Throttle internal power stage is deemed shorted high or low through internal hardware checks which commence when throttle control PWM duty is excessive ▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined failures where some circuit quantity, reported through serial data, is above a specified range ▪ Electric throttle control circuits short circuit to power, short circuit to ground ▪ Electric throttle actuator control power and ground circuits short circuit to each another ▪ Electric throttle actuator failure	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Battery voltage greater than 8 volts ▪ No throttle driver error present ▪ Throttle adaptions completed ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator circuits for short circuit to power, short circuit to ground ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator power and ground circuits for short circuit to each other ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected ▪ Clear the DTC and retest ▪ Check and install a new electric throttle actuator as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2108-00	Throttle Actuator "A" Control Module Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_TVMPOS -
		  ▪ Circuit reference - O_T_TVMNEG -
		▪ Monitor Description
		  ▪ Throttle internal power stage is deemed over current through internal hardware checks which commence when throttle control PWM duty is excessive
		▪ Prioritized List of Possible Causes
		  ▪ Electric throttle control circuits short circuit to power, short circuit to ground
		  ▪ Electric throttle actuator control power and ground circuits short circuit to each another
		  ▪ Electric throttle actuator failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Battery voltage greater than 8 volts
			  ▪ No throttle driver error present
			  ▪ Throttle adaptions completed
			  ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator circuits for short circuit to power, short circuit to ground
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator power and ground circuits for short circuit to each other
			  ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new electric throttle actuator as required

P2111-00	Throttle Actuator "A" Control System - Stuck Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_TVMPOS -
		  ▪ Circuit reference - O_T_TVMNEG -
		▪ Monitor Description
		  ▪ Throttle control PWM -ve duty is excessive in blade closing direction
		▪ Prioritized List of Possible Causes
		  ▪ Other electric throttle unit related Diagnostic Trouble Code(s)
		  ▪ Electric throttle control circuits short circuit to power, short circuit to ground
		  ▪ Electric throttle actuator control power and ground circuits short circuit to each another
		  ▪ Electric throttle actuator failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Battery voltage greater than 8 volts
			  ▪ No throttle driver error present
			  ▪ Throttle adaptions completed
			  ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator circuits for short circuit to power, short circuit to ground
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator power and ground circuits for short circuit to each other
			  ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new electric throttle actuator as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2112-85	Throttle Actuator "A" Control System - Stuck Closed - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_TVMPOS -
		  ▪ Circuit reference - O_T_TVMNEG -
		▪ Monitor Description
		  ▪ Throttle control PWM +ve duty is excessive in blade closing direction
		▪ Prioritized List of Possible Causes
		  ▪ Other electric throttle unit related Diagnostic Trouble Code(s)
		  ▪ Electric throttle control circuits short circuit to power, short circuit to ground
		  ▪ Electric throttle actuator control power and ground circuits short circuit to each another
		  ▪ Electric throttle actuator failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Battery voltage greater than 8 volts
			  ▪ No throttle driver error present
			  ▪ Throttle adaptions completed
			  ▪ Throttle powered with Ignition on (no other throttle Diagnostic Trouble Code(s) present) for a period greater than 30 seconds. To make sure that a fault that is only present at high throttle
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator circuits for short circuit to power, short circuit to ground
			  ▪ Refer to the electrical circuit diagrams and check the electric throttle actuator power and ground circuits for short circuit to each other
			  ▪ Refer to the electrical circuit diagrams and check for short circuit across electric throttle power and ground connections with harness disconnected
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new electric throttle actuator as required

P2119-00	Throttle Actuator "A" Control Throttle Body Range/Performance - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ During ECU power up the throttle spring checks have been performed the throttle power stage is disabled and the return spring should mechanically close the blade. If the blade does not close the return spring has failed
		▪ Prioritized List of Possible Causes
		  ▪ Other related Diagnostic Trouble Code(s)
		  ▪ Electric throttle return spring failure
		  ▪ Electric throttle blocked or restricted
		  ▪ Electric throttle actuator failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, spring check will be performed if blade does not achieve a position close to the learnt limp home position a fault is set
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first
			  ▪ Check electric throttle return spring for correct operation
			  ▪ Check electric throttle for blockages or obstruction that prevent correct movement
			  ▪ Clear the DTC and retest
			  ▪ Check and install a new electric throttle actuator as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2119-92	Throttle Actuator "A" Control Throttle Body Range/Performance - Performance or incorrect operation	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ When the throttle has been requested to attain its limp home position should the actual position deviate in the open direction this could result in the hazard of unintended acceleration without this fault detection and application of a severe fuel restriction
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the component performance is outside its expected range or operating in an incorrect way
		  ▪ During powertrain control module power up the learnt limp home position is outside of specification
		  ▪ Throttle butterfly has been unable to move due to electrical trouble. If the cause is electrical other electric throttle unit related Diagnostic Trouble Code(s) will be set
		  ▪ Throttle butterfly unable to move due to obstruction
		  ▪ Throttle butterfly mechanically stuck
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, limp home position check will be performed if blade does not achieve a position close to the learnt limp home position a fault is set
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Repair wiring harness as required. Clear the DTC and retest
			  ▪ Check there is not any obstruction preventing electric throttle butterfly movement
			  ▪ If the DTC resets check and install a new electric throttle unit as required

P2119-97	Throttle Actuator "A" Control Throttle Body Range/Performance - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ During ECU power up the throttle spring checks have been performed and the throttle was unable to open the throttle blade prior to checking the return spring, If the fault is electrical other Throttle electrical Diagnostic Trouble Codes will be present and should be rectified prior to investigating possibility that the throttle blade is unable to move when requested due to mechanical failure/physical obstruction internally
		▪ Prioritized List of Possible Causes
		  ▪ The mechanical throttle butterfly spring has been unable to open the throttle butterfly
		  ▪ Throttle butterfly has been unable to close due to electrical trouble. If the cause is electrical other electric throttle unit related Diagnostic Trouble Code(s) will be set
		  ▪ Throttle butterfly unable to move due to obstruction
		  ▪ Electric throttle unit failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, spring check will be performed if blade does not achieve opening position of 15% beyond the learnt limp home position a fault is set
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Repair wiring harness as required. Clear the DTC and retest
			  ▪ Check there is not any obstruction preventing electric throttle butterfly movement
			  ▪ If the DTC resets check and install a new electric throttle unit as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2122-00	Throttle/Pedal Position Sensor/Switch "D" Circuit Low - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference - I_A_APP1 -
		- Monitor Description
		  - Raw accelerator pedal voltage "APP1" out of range low
		- Prioritized List of Possible Causes
		  - Harness failure - Accelerator pedal position sensor circuit
		  - Accelerator pedal position sensor failure
			- Vehicle Conditions to enable DTC Logging strategy
			  - Ignition on 5 seconds, no regulated power supply errors
			- Prioritized Checks to Perform
			  - This DTC is set when the powertrain control module detects the pedal demand 1 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  - Check and install a new accelerator pedal position sensor as required

P2123-00	Throttle/Pedal Position Sensor/Switch "D" Circuit High - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference - I_A_APP1 -
		- Monitor Description
		  - Raw accelerator pedal voltage "APP1" out of range high
		- Prioritized List of Possible Causes
		  - Harness failure - Accelerator pedal position sensor circuit
		  - Accelerator pedal position sensor failure
			- Vehicle Conditions to enable DTC Logging strategy
			  - Ignition on 5 seconds, no regulated power supply errors
			- Prioritized Checks to Perform
			  - This DTC is set when the powertrain control module detects the pedal demand 1 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  - Check and install a new accelerator pedal position sensor as required

P2127-00	Throttle/Pedal Position Sensor/Switch "E" Circuit Low - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference - I_A_APP2 -
		- Monitor Description
		  - Raw accelerator pedal voltage "APP2" out of range low
		- Prioritized List of Possible Causes
		  - Harness failure - Accelerator pedal position sensor circuit
		  - Accelerator pedal position sensor failure
			- Vehicle Conditions to enable DTC Logging strategy
			  - Ignition on 5 seconds, no regulated power supply errors
			- Prioritized Checks to Perform
			  - This DTC is set when the powertrain control module detects the pedal demand 2 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  - Check and install a new accelerator pedal position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2128-00	Throttle/Pedal Position Sensor/Switch "E" Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_APP2 -
		▪ Monitor Description
		  ▪ Raw accelerator pedal voltage "APP2" out of range high
		▪ Prioritized List of Possible Causes
		  ▪ Harness failure - Accelerator pedal position sensor circuit
		  ▪ Accelerator pedal position sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on 5 seconds, no regulated power supply errors
			▪ Prioritized Checks to Perform
			  ▪ This DTC is set when the powertrain control module detects the pedal demand 2 signal range is high. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new accelerator pedal position sensor as required

P2135-00	Throttle/Pedal Position Sensor/Switch "A"/"B" Voltage Correlation - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_TPS2 -
		  ▪ Circuit reference - I_A_TPS1 -
		▪ Prioritized List of Possible Causes
		  ▪ Multiple throttle position failures could be present
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Make sure all throttle position circuit faults have been rectified prior to investigating this fault. Engine running at idle for 5 seconds, also perform a short less than 10 minute varied drive cycle to make sure fault has been rectified, make sure light and high engine loads are included
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check for related Diagnostic Trouble Code(s) and refer to the relevant DTC index, clear the DTC and retest the system

P2138-00	Throttle/Pedal Position Sensor/Switch "D"/"E" Voltage Correlation - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_APP1 -
		  ▪ Circuit reference - I_A_APP2 -
		▪ Monitor Description
		  ▪ Excessive difference between raw voltage of APP1 and APP2
		▪ Prioritized List of Possible Causes
		  ▪ Harness failure - Accelerator pedal position sensor circuit
		  ▪ Accelerator pedal position sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on 5 seconds
			▪ Prioritized Checks to Perform
			  ▪ This DTC is set when the powertrain control module detects an excessive difference between the pedal demand signal 1 and signal 2. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensors. Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new accelerator pedal position sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2162-26	Vehicle Speed Sensor "A"/"B" Correlation - Signal rate of change below threshold	- Electrical Cause - Yes - Mechanical Cause - No - Monitor Description - Compares ABS Vehicle speed to TCM vehicle speed - Prioritized List of Possible Causes - Vehicle speed comparison difference between transmission control module and anti-lock brake system control module	- Vehicle Conditions to enable DTC Logging strategy - Start engine and drive at 30 mph for 1 minute - Prioritized Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
P2162-27	Vehicle Speed Sensor "A"/"B" Correlation - Signal rate of change above threshold	- Electrical Cause - Yes - Mechanical Cause - No - Monitor Description - Compares TCM Vehicle speed to ABS vehicle speed - Prioritized List of Possible Causes - Vehicle speed comparison difference between transmission control module and anti-lock brake system control module	- Vehicle Conditions to enable DTC Logging strategy - Start engine and drive at 30 mph for 1 minute - Prioritized Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module - Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2169-13	Exhaust Pressure Regulator Vent Solenoid Control Circuit/Open - Circuit open	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_S_EXPL - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Active exhaust actuator power or ground circuit open circuit, high resistance ▪ Active exhaust actuator control circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Active exhaust actuator or valve seized ▪ Active exhaust actuator failure	**NOTES:** ▪ On some vehicles, movement of the active exhaust valve is observable through the left tailpipe. ▪ For Jaguar F-Type only, the active exhaust switch should flash 8 times when operated. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Start engine, increase engine speed to 4000 rpm for 2 seconds, return to idle speed ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the active exhaust actuator power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the active exhaust actuator control circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Remove the active exhaust actuator. Check the active exhaust valve for seizure. Rectify as necessary ▪ Check and install a new active exhaust actuator only when diagnosed as failed

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P2170-11</td>
<td>Exhaust Pressure Regulator Vent Solenoid Control Circuit Low - Circuit short to ground</td>
<td>
Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - O_S_EXPL -
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Active exhaust actuator power or ground circuit open circuit, high resistance
Active exhaust actuator control circuit short circuit to ground, short circuit to power, open circuit, high resistance
Active exhaust actuator or valve seized
Active exhaust actuator failure

</td>
<td>NOTES:

On some vehicles, movement of the active exhaust valve is observable through the left tailpipe.
For Jaguar F-Type only, the active exhaust switch should flash 8 times when operated.

Vehicle Conditions to Enable DTC Logging Strategy
Start engine, increase engine speed to 4000 rpm for 2 seconds, return to idle speed
Prioritized Checks to Perform
Inspect connectors for signs of water ingress and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the active exhaust actuator power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
Refer to the electrical circuit diagrams and check the active exhaust actuator control circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
Remove the active exhaust actuator. Check the active exhaust valve for seizure. Rectify as necessary
Check and install a new active exhaust actuator only when diagnosed as failed

</td>
</tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2170-4B	Exhaust Pressure Regulator Vent Solenoid Control Circuit Low - Over temperature	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_S_EXPL -
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Active exhaust actuator power or ground circuit open circuit, high resistance
		  - Active exhaust actuator control circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Active exhaust actuator or valve seized
		  - Active exhaust actuator failure
			NOTES:
			- On some vehicles, movement of the active exhaust valve is observable through the left tailpipe.
			- For Jaguar F-Type only, the active exhaust switch should flash 8 times when operated.
			
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Start engine, increase engine speed to 4000 rpm for 2 seconds, return to idle speed
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check the active exhaust actuator power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  - Refer to the electrical circuit diagrams and check the active exhaust actuator control circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Remove the active exhaust actuator. Check the active exhaust valve for seizure. Rectify as necessary
			  - Check and install a new active exhaust actuator only when diagnosed as failed

P2171-12	Exhaust Pressure Regulator Vent Solenoid Control Circuit high - Circuit short to battery	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - O_S_EXPL -
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Active exhaust solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Active exhaust solenoid failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Start engine then press loud button and look for the Button LED flashing
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check the active exhaust solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Check and install a new active exhaust solenoid only when diagnosed as failed
			  - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2176-00	Throttle Actuator "A" Control System - Idle Position Not Learned - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Prior valid EEPROM stored throttle adaption data is different to that calculated on this ECU power up, a replacement throttle has been installed
		▪ Prioritized List of Possible Causes
		  ▪ Electric throttle actuator adaption routine not completed
		  ▪ Replacement throttle installed but adaption routine not completed
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, adaption will be performed, if unsuccessful relevant Diagnostic Trouble Codes shall be set allowing specific rectification action to be performed
			▪ Prioritized Checks to Perform
			  ▪ Allow powertrain control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2176-51	Throttle Actuator "A" Control System - Idle Position Not Learned - Not programed	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Vehicle operating voltage is too low to allow new adaptions to be performed
		▪ Prioritized List of Possible Causes
		  ▪ Voltage too low to perform electric throttle actuator adaption routine
		  ▪ Electric throttle actuator adaption routine not completed
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, adaption will be performed if entry conditions are present
			▪ Prioritized Checks to Perform
			  ▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual
			  ▪ Allow powertrain control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2176-52	Throttle Actuator "A" Control System - Idle Position Not Learned - Not activated	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Vehicle operating conditions and environmental conditions are incorrect to allow new adaptions to be performed
		▪ Prioritized List of Possible Causes
		  ▪ Electric throttle actuator adaption routine entry conditions are not met
		  ▪ Electric throttle actuator adaption routine not completed
		  ▪ Replacement throttle installed but adaption routine not completed
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition off - On and wait 60 seconds prior to starting engine, adaption will be performed if entry conditions are present
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first
			  ▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual
			  ▪ Allow powertrain control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2176-54	Throttle Actuator "A" Control System - Idle Position Not Learned - Missing calibration	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ EEPROM throttle end stop or limp home adaption data is not present for throttle and new adaption data this trip is also out of valid voltage range ▪ Prioritized List of Possible Causes ▪ Electric throttle actuator adaption routine entry conditions are not met ▪ Electric throttle actuator adaption routine not completed ▪ Replacement throttle installed but adaption routine not completed ▪ Electric throttle actuator failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition off - On and wait 60 seconds prior to starting engine, adaption will be performed, if unsuccessful relevant Diagnostic Trouble Codes shall be set allowing specific rectification action to be performed ▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Check battery is in fully charged and in a serviceable condition using a battery tester and battery care manual ▪ Allow powertrain control module to power down. Turn on ignition and wait for 60 seconds. Start engine, this will allow new throttle adaption data to be stored ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest ▪ Check and install a new electric throttle actuator as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2177-00	System Too Lean Off Idle Bank 1 - No sub type information	▪ Air intake system leakage
		▪ Exhaust system leakage
		▪ High ethanol content fuel (E85)
		  ▪ Using E85 on a gasoline engine will result in a trim of approximately 50% under all operating conditions. A mix of E85 and gasoline will result in trims somewhere between 0 and 50% depending on the mix
		▪ Vacuum leak
		▪ Mass air flow sensor contamination
		▪ Incorrect fuel pressure
			NOTE:
			Operational requirements needed to allow the monitor to be fully tested. Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position)
			▪ Using the Jaguar Land Rover approved smoke test system, check for air intake system leakage
			▪ Using the Jaguar Land Rover approved smoke test system, check for exhaust system leakage
			▪ Using the Jaguar Land Rover approved diagnostic equipment check fuel trims at idle and under load
			  ▪ If fuel trims (Long and Short combined) are significantly better under load, check for a vacuum leak
			  ▪ If the fuel trims are consistent across the range (about +45% for straight E85; or less if it's a mix). E85 could be a potential cause
			  ▪ Take a sample of fuel and check if it smells odd. If it smells normal complete an ethanol test. If the fuel is not gasoline drain and re fill the tank
			  ▪ **Ethanol test.** Mix a few inches of water and a few inches of fuel in a bottle and shake them together; the ethanol will bind with the water and come out of the gasoline. You will be left with what appears to be more water in the bottle than you started with and the gasoline floating on top. The amount of gasoline that has 'changed' to water is the amount of ethanol that was in the fuel
			▪ Check for mass air flow sensor contamination
			▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2178-00	System Too Rich Off Idle Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Detects when primary fueling adaption multiplier is too small
		▪ Prioritized List of Possible Causes
		  ▪ Blocked air filter
		  ▪ Air intake system restriction
		  ▪ Mass air flow sensor contamination
		  ▪ Exhaust system restriction
		  ▪ Incorrect fuel pressure
			NOTE:
			This DTC may set as a result of P0191-85 setting. Complete the repair actions for P0191-85 first, clear the Diagnostic Trouble Code(s) and retest. If this DTC returns, perform the checks below
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 10 minutes with several periods of steady operation for more than 20 seconds (no gear shifts, steady accelerator position)
			▪ Prioritized Checks to Perform
			  ▪ Check for blocked air filter
			  ▪ Check for air intake system restriction
			  ▪ Check for mass air flow sensor contamination
			  ▪ Check for exhaust system restriction
			  ▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2183-26	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal rate of change below threshold	▪ Purpose of the DTC
		  ▪ To detect an implausible radiator outlet temperature sensor signal
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RADWTS -
		  ▪ Circuit reference - G_R_RADWTS -
		▪ Prioritized List of Possible Causes
		  ▪ Cooling system blocked or restricted
		  ▪ Icing in the coolant due to incorrect antifreeze mixture
		  ▪ Radiator outlet temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance
		  ▪ Radiator outlet temperature sensor failure
			NOTE:
			When the engine is cold, the engine coolant temperature sensor and radiator outlet temperature sensor signal values should not differ by greater than 10°C.
			▪ Prioritized Checks to Perform
			  ▪ Check the cooling system for blockages or restrictions. Rectify as necessary
			  ▪ Check the specific gravity of the coolant. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check radiator outlet temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance. Repair the wiring harness or install a new radiator outlet temperature sensor as necessary
			  ▪ Install a new radiator outlet temperature sensor

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2183-62	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal compare failure	▪ Prioritized List of Possible Causes ▪ Blockage in coolant system ▪ Insufficient mixture of antifreeze creating possibility of icing in coolant system ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Engine coolant temperature sensor failure	▪ Prioritized Checks to Perform ▪ Check coolant system for blockages or restrictions that prevent coolant flow ▪ Check for the correct level of antifreeze within the coolant ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check and install a new engine coolant temperature sensor as required
P2183-64	Engine Coolant Temperature Sensor 2 Circuit Range/Performance - Signal plausibility failure	▪ Prioritized List of Possible Causes ▪ Blockage in coolant system ▪ Insufficient mixture of antifreeze creating possibility of icing in coolant system ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Engine coolant temperature sensor circuit short circuit to power, short circuit to ground, open circuit, high resistance ▪ Engine coolant temperature sensor failure	▪ Prioritized Checks to Perform ▪ Check coolant system for blockages or restrictions that prevent coolant flow ▪ Check for the correct level of antifreeze within the coolant ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor for short circuit to power, short circuit to ground, open circuit, high resistance ▪ Check and install a new engine coolant temperature sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2184-00	Engine Coolant Temperature Sensor 2 Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RADWTS -
		  ▪ Circuit reference - G_R_RADWTS -
		▪ Monitor Description
		  ▪ Manifold TMAP sensor temperature sensor circuit or short circuit to battery or 5 volts
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Radiator outlet temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Radiator outlet temperature sensor failure
		  ▪ Engine coolant temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on or engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the radiator outlet temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new radiator outlet temperature sensor only when diagnosed as failed
			  ▪ Check and install a new engine coolant temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2185-00	Engine Coolant Temperature Sensor 2 Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_RADWTS -
		  ▪ Circuit reference - G_R_RADWTS -
		▪ Monitor Description
		  ▪ Manifold TMAP sensor temperature sensor circuit or short circuit to battery or 5 volts
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Radiator outlet temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Radiator outlet temperature sensor failure
		  ▪ Engine coolant temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on or engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the radiator outlet temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new radiator outlet temperature sensor only when diagnosed as failed
			  ▪ Check and install a new engine coolant temperature sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2187-00	System Too Lean at Idle Bank 1 - No sub type information	▪ Air intake system leakage
		▪ Exhaust system leakage
		▪ High ethanol content fuel (E85)
		  ▪ Using E85 on a gasoline engine will result in a trim of approximately 50% under all operating conditions. A mix of E85 and gasoline will result in trims somewhere between 0 and 50% depending on the mix
		▪ Vacuum leak
		▪ Pipe detached/union leakage between intake manifold and cylinder head
		▪ Fuel pressure too low
		▪ Mass air flow sensor contaminated
		▪ Purge valve internal failure
			NOTE:
			This DTC may set as a result of P0191-85 setting. Complete the repair actions for P0191-85 first, clear the Diagnostic Trouble Code(s) and retest. If this DTC returns, perform the checks below
			▪ Using the Jaguar Land Rover approved smoke test system, check for air intake system leakage
			▪ Using the Jaguar Land Rover approved smoke test system, check for exhaust system leakage
			▪ Using the Jaguar Land Rover approved diagnostic equipment check fuel trims at idle and under load
			  ▪ If fuel trims (Long and Short combined) are significantly better under load, check for a vacuum leak
			  ▪ If the fuel trims are consistent across the range (about +45% for straight E85; or less if it's a mix). E85 could be a potential cause
			  ▪ Take a sample of fuel and check if it smells odd. If it smells normal complete an ethanol test. If the fuel is not gasoline drain and re fill the tank
			  ▪ **Ethanol test.** Mix a few inches of water and a few inches of fuel in a bottle and shake them together; the ethanol will bind with the water and come out of the gasoline. You will be left with what appears to be more water in the bottle than you started with and the gasoline floating on top. The amount of gasoline that has 'changed' to water is the amount of ethanol that was in the fuel
			▪ Check intake manifold to cylinder head union for air leaks or pipe disconnected
			▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification
			▪ Check for mass air flow sensor contamination
			▪ Check all heated oxygen sensors are installed correctly
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			▪ If the fault persists, install a new purge valve when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2188-00	System Too Rich at Idle Bank 1 - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Detects when primary fueling adaption offset is too small
		▪ Prioritized List of Possible Causes
		  ▪ Air intake system leakage
		  ▪ Pipe detached/union leakage between intake manifold and cylinder head
		  ▪ Fuel pressure too low
		  ▪ Mass air flow sensor contaminated
		  ▪ Exhaust system leakage
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle in Drive
			▪ Prioritized Checks to Perform
			  ▪ Check for air intake system leakage
			  ▪ Check intake manifold to cylinder head union for air leaks or pipe disconnected
			  ▪ Refer to the relevant section of the workshop manual and check fuel pressure is to specification
			  ▪ Check for mass air flow sensor contamination
			  ▪ Check for exhaust system leakage
			  ▪ Check all heated oxygen sensors are installed correctly
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219C-66	Cylinder 1 Air-Fuel Ratio Imbalance - Signal has too many transitions/events	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Engine misfire
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Spark plug installation
		  ▪ Spark plug condition
		  ▪ Fuel injector seal leaking
		  ▪ Intake manifold seal leaking
		  ▪ Fuel injector blocked, damaged, internal failure
		  ▪ Fuel rail blocked, damaged
		  ▪ Ignition coil circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ignition coil internal failure
		  ▪ Continuously variable valve lift system fault
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200 to 2400 rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for engine misfire related Diagnostic Trouble Code(s) and perform the relevant corrective actions
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the spark plug for correct installation. Rectify as necessary
			  ▪ Check the condition of the spark plug. Install a new spark plug as necessary
			  ▪ Check the fuel injector seal for leaks. Rectify as necessary
			  ▪ Check the intake manifold seal for leaks. Rectify as necessary
			  ▪ Check the integrity of the fuel injector. Install a new fuel injector as necessary
			  ▪ Check the integrity of the fuel rail. Install a new fuel rail as necessary
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the integrity of the ignition coil. Install a new ignition coil as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for continuously variable valve lift related Diagnostic Trouble Code(s) and perform the relevant corrective actions

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219C-84	Cylinder 1 Air-Fuel Ratio Imbalance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) below threshold, (richer mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Blockage or restriction of air entering cylinder
		  ▪ Restriction of fuel rail
		  ▪ Blockage or restriction within exhaust manifold
		  ▪ Camshaft profile switching system failure
		  ▪ Fuel injector
		  ▪ Friction in valve train components
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Check for blockage or restriction of air entering cylinder
			  ▪ Check for restriction of fuel rail
			  ▪ Check for blockage or restriction within exhaust manifold
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P219C-85	Cylinder 1 Air-Fuel Ratio Imbalance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) above threshold, (leaner mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leak on intake manifold-to-cylinder head connection
		  ▪ External fuel leak at fuel injector
		  ▪ Fuel restriction within fuel injector
		  ▪ Restriction of fuel rail
		  ▪ Air leak around spark plug
		  ▪ Air leak around fuel injector
		  ▪ Fuel injector
		  ▪ Ignition coil
		  ▪ Camshaft profile switching system failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check for air leak on intake manifold-to-cylinder head connection
			  ▪ Check for external fuel leak at fuel injector
			  ▪ Check for fuel restriction within fuel injector
			  ▪ Check for restriction of fuel rail
			  ▪ Check for air leak around spark plug
			  ▪ Check for air leak around fuel injector
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219D-66	Cylinder 2 Air-Fuel Ratio Imbalance - Signal has too many transitions/events	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Engine misfire
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Spark plug installation
		  ▪ Spark plug condition
		  ▪ Fuel injector seal leaking
		  ▪ Intake manifold seal leaking
		  ▪ Fuel injector blocked, damaged, internal failure
		  ▪ Fuel rail blocked, damaged
		  ▪ Ignition coil circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ignition coil internal failure
		  ▪ Continuously variable valve lift system fault
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200 to 2400 rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for engine misfire related Diagnostic Trouble Code(s) and perform the relevant corrective actions
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the spark plug for correct installation. Rectify as necessary
			  ▪ Check the condition of the spark plug. Install a new spark plug as necessary
			  ▪ Check the fuel injector seal for leaks. Rectify as necessary
			  ▪ Check the intake manifold seal for leaks. Rectify as necessary
			  ▪ Check the integrity of the fuel injector. Install a new fuel injector as necessary
			  ▪ Check the integrity of the fuel rail. Install a new fuel rail as necessary
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the integrity of the ignition coil. Install a new ignition coil as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for continuously variable valve lift related Diagnostic Trouble Code(s) and perform the relevant corrective actions

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219D-84	Cylinder 2 Air-Fuel Ratio Imbalance - Signal below allowable range	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Monitor Description
		  - Inferred mixture strength (Lambda) below threshold, (richer mixture)
		- Prioritized List of Possible Causes
		  - Blockage or restriction of air entering cylinder
		  - Restriction of fuel rail
		  - Blockage or restriction within exhaust manifold
		  - Camshaft profile switching system failure
		  - Fuel injector
		  - Friction in valve train components
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Engine coolant temperature greater than 50°C
			  - Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			- Prioritized Checks to Perform
			  - Check for blockage or restriction of air entering cylinder
			  - Check for restriction of fuel rail
			  - Check for blockage or restriction within exhaust manifold
			  - Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  - Check and install a new fuel injector only when diagnosed as failed
			  - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P219D-85	Cylinder 2 Air-Fuel Ratio Imbalance - Signal above allowable range	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Monitor Description
		  - Inferred mixture strength (Lambda) above threshold, (leaner mixture)
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  - Air leak on intake manifold-to-cylinder head connection
		  - External fuel leak at fuel injector
		  - Fuel restriction within fuel injector
		  - Restriction of fuel rail
		  - Air leak around spark plug
		  - Air leak around fuel injector
		  - Fuel injector
		  - Ignition coil
		  - Camshaft profile switching system failure
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Engine coolant temperature greater than 50°C
			  - Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  - Check for air leak on intake manifold-to-cylinder head connection
			  - Check for external fuel leak at fuel injector
			  - Check for fuel restriction within fuel injector
			  - Check for restriction of fuel rail
			  - Check for air leak around spark plug
			  - Check for air leak around fuel injector
			  - Check and install a new fuel injector only when diagnosed as failed
			  - Check and install a new ignition coil only when diagnosed as failed
			  - Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219E-66	Cylinder 3 Air-Fuel Ratio Imbalance - Signal has too many transitions/events	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Engine misfire
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Spark plug installation
		  ▪ Spark plug condition
		  ▪ Fuel injector seal leaking
		  ▪ Intake manifold seal leaking
		  ▪ Fuel injector blocked, damaged, internal failure
		  ▪ Fuel rail blocked, damaged
		  ▪ Ignition coil circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Ignition coil internal failure
		  ▪ Continuously variable valve lift system fault
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200 to 2400 rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for engine misfire related Diagnostic Trouble Code(s) and perform the relevant corrective actions
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the spark plug for correct installation. Rectify as necessary
			  ▪ Check the condition of the spark plug. Install a new spark plug as necessary
			  ▪ Check the fuel injector seal for leaks. Rectify as necessary
			  ▪ Check the intake manifold seal for leaks. Rectify as necessary
			  ▪ Check the integrity of the fuel injector. Install a new fuel injector as necessary
			  ▪ Check the integrity of the fuel rail. Install a new fuel rail as necessary
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check the integrity of the ignition coil. Install a new ignition coil as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for continuously variable valve lift related Diagnostic Trouble Code(s) and perform the relevant corrective actions

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219E-84	Cylinder 3 Air-Fuel Ratio Imbalance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) below threshold, (richer mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Blockage or restriction of air entering cylinder
		  ▪ Restriction of fuel rail
		  ▪ Blockage or restriction within exhaust manifold
		  ▪ Camshaft profile switching system failure
		  ▪ Fuel injector
		  ▪ Friction in valve train components
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Check for blockage or restriction of air entering cylinder
			  ▪ Check for restriction of fuel rail
			  ▪ Check for blockage or restriction within exhaust manifold
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P219E-85	Cylinder 3 Air-Fuel Ratio Imbalance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) above threshold, (leaner mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leak on intake manifold-to-cylinder head connection
		  ▪ External fuel leak at fuel injector
		  ▪ Fuel restriction within fuel injector
		  ▪ Restriction of fuel rail
		  ▪ Air leak around spark plug
		  ▪ Air leak around fuel injector
		  ▪ Fuel injector
		  ▪ Ignition coil
		  ▪ Camshaft profile switching system failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check for air leak on intake manifold-to-cylinder head connection
			  ▪ Check for external fuel leak at fuel injector
			  ▪ Check for fuel restriction within fuel injector
			  ▪ Check for restriction of fuel rail
			  ▪ Check for air leak around spark plug
			  ▪ Check for air leak around fuel injector
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P219F-66	Cylinder 4 Air-Fuel Ratio Imbalance - Signal has too many transitions/events	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ Engine misfire ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Spark plug installation ▪ Spark plug condition ▪ Fuel injector seal leaking ▪ Intake manifold seal leaking ▪ Fuel injector blocked, damaged, internal failure ▪ Fuel rail blocked, damaged ▪ Ignition coil circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Ignition coil internal failure ▪ Continuously variable valve lift system fault	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine coolant temperature greater than 50°C ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200 to 2400 rpm, maintain these conditions for 10 minutes (highway cruise) ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for engine misfire related Diagnostic Trouble Code(s) and perform the relevant corrective actions ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check the spark plug for correct installation. Rectify as necessary ▪ Check the condition of the spark plug. Install a new spark plug as necessary ▪ Check the fuel injector seal for leaks. Rectify as necessary ▪ Check the intake manifold seal for leaks. Rectify as necessary ▪ Check the integrity of the fuel injector. Install a new fuel injector as necessary ▪ Check the integrity of the fuel rail. Install a new fuel rail as necessary ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check the integrity of the ignition coil. Install a new ignition coil as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for continuously variable valve lift related Diagnostic Trouble Code(s) and perform the relevant corrective actions

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P219F-84	Cylinder 4 Air-Fuel Ratio Imbalance - Signal below allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) below threshold, (richer mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Blockage or restriction of air entering cylinder
		  ▪ Restriction of fuel rail
		  ▪ Blockage or restriction within exhaust manifold
		  ▪ Camshaft profile switching system failure
		  ▪ Fuel injector
		  ▪ Friction in valve train components
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Check for blockage or restriction of air entering cylinder
			  ▪ Check for restriction of fuel rail
			  ▪ Check for blockage or restriction within exhaust manifold
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P219F-85	Cylinder 4 Air-Fuel Ratio Imbalance - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Inferred mixture strength (Lambda) above threshold, (leaner mixture)
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leak on intake manifold-to-cylinder head connection
		  ▪ External fuel leak at fuel injector
		  ▪ Fuel restriction within fuel injector
		  ▪ Restriction of fuel rail
		  ▪ Air leak around spark plug
		  ▪ Air leak around fuel injector
		  ▪ Fuel injector
		  ▪ Ignition coil
		  ▪ Camshaft profile switching system failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine coolant temperature greater than 50°C
			  ▪ Drive vehicle at steady load in 6th gear to achieve an engine speed of 1200-2400rpm, maintain these conditions for 10 minutes (highway cruise)
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check for air leak on intake manifold-to-cylinder head connection
			  ▪ Check for external fuel leak at fuel injector
			  ▪ Check for fuel restriction within fuel injector
			  ▪ Check for restriction of fuel rail
			  ▪ Check for air leak around spark plug
			  ▪ Check for air leak around fuel injector
			  ▪ Check and install a new fuel injector only when diagnosed as failed
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Refer to the relevant section of the workshop manual and check the camshaft profile switching system for correct operation
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2227-23	Barometric Pressure Sensor "A" Circuit Range/Performance - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Compare the ambient pressure sensor value, +/- sensor tolerance, to maximum and minimum thresholds, derived from the total mean value of all the pressure sensors during engine start
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition must have been switched off for greater than or equal to 5 seconds. Monitor enabled during cranking
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P2227-24	Barometric Pressure Sensor "A" Circuit Range/Performance - Signal stuck high	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Compare the ambient pressure sensor value, +/- sensor tolerance, to maximum and minimum thresholds, derived from the total mean value of all the pressure sensors during engine start
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition must have been switched off for greater than or equal to 5 seconds. Monitor enabled during cranking
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P2227-26	Barometric Pressure Sensor "A" Circuit Range/Performance - Signal rate of change below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Compare raw ambient pressure value to a filtered ambient pressure value, fault is detected if the raw value deviates away from the filtered value greater than threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Continuously monitored
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2227-27	Barometric Pressure Sensor "A" Circuit Range/Performance - Signal rate of change above threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Compare raw ambient pressure value to a filtered ambient pressure value, fault is detected if the raw value deviates away from the filtered value greater than threshold ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Continuously monitored ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P2227-49	Barometric Pressure Sensor "A" Circuit Range/Performance - Internal electronic failure	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Sensor chip internal fault check communicated to micro controller through SPI bus ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Continuously monitored ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
P2228-16	Barometric Pressure Sensor "A" Circuit Low - Circuit voltage below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Internal pressure sensor fault. Sensor shorted to internal ground ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, start engine and idle for 30 seconds ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2229-17	Barometric Pressure Sensor "A" Circuit High - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Internal pressure sensor fault. Sensor shorted to internal battery supply
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Powertrain control module failure. The sensor is installed within the powertrain control module and is not serviceable
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, start engine and idle for 30 seconds
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check and install a new powertrain control module only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module

P2237-00	O2 Sensor Positive Current Control Circuit/Open Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSCP -
		▪ Monitor Description
		  ▪ Detects open circuit of the sensor circuit lines
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor circuit open circuit, high resistance
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new heated oxygen sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2243-00	O2 Sensor Reference Voltage Circuit/Open Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_LSVN -
		▪ Monitor Description
		  ▪ Detects open circuit of the sensor circuit lines
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor circuit open circuit, high resistance
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new heated oxygen sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2251-00	O2 Sensor Negative Current Control Circuit/Open Bank 1 Sensor 1 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_R_LSVG -
		▪ Monitor Description
		  ▪ Detects open circuit of the sensor circuit lines
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Heated oxygen sensor circuit open circuit, high resistance
		  ▪ Heated oxygen sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Start engine and allow to idle for 5 minutes
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuit for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new heated oxygen sensor only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2261-73	Turbocharger/Supercharger Bypass Valve "A" - Mechanical - Actuator stuck closed	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Plausibility error
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Compressor bypass control actuator - Wiring integrity
		  ▪ Compressor bypass control actuator - Mechanical integrity
		  ▪ Compressor bypass control actuator failure
		▪ Turbocharger recirculation valve mechanical integrity
		▪ Turbocharger recirculation valve failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of wastegate actuation
			▪ Check turbocharger wastegate voltage at closed position
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Check the mechanical integrity of the turbocharger recirculation valve. Repair or replace as necessary
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P226D-00	Particulate Filter Deteriorated/Missing Substrate Bank 1 - No sub type information	▪ MIL ▪ Yes ▪ Prioritized List of Possible Causes ▪ Other gasoline particulate filter related Diagnostic Trouble Code(s) ▪ Other oxygen sensor related Diagnostic Trouble Code(s) ▪ Gasoline particulate filter substrate is missing ▪ Gasoline particulate filter severely damaged ▪ Gasoline particulate filter failure	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the powertrain control module for related GPF and oxygen sensor Diagnostic Trouble Code(s). Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure ▪ Remove the GPF and visually inspect for severe damage and correct installation (if available, use camera/boroscope to inspect front and rear face of GPF through oxygen sensor boss both upstream and downstream of GPF ▪ Install a new GPF, only if diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2270-00	O2 Sensor Signal Biased/Stuck Lean Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Activity of the sensor is too low
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leakage between heated oxygen sensor and catalyst
		  ▪ Heated oxygen sensor tip damaged
		  ▪ Heated oxygen sensor tip blocked
		  ▪ Heated oxygen sensor tip contaminated by excessive oil consumption
		  ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 15 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Check heated oxygen sensor for tip damage
			  ▪ Check heated oxygen sensor for tip blocked
			  ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Complete oil consumption check
			  ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2271-00	O2 Sensor Signal Biased/Stuck Rich Bank 1 Sensor 2 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Activity of the sensor is too low
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leakage between heated oxygen sensor and catalyst
		  ▪ Heated oxygen sensor tip damaged
		  ▪ Heated oxygen sensor tip blocked
		  ▪ Heated oxygen sensor tip contaminated by excessive oil consumption
		  ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 15 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Check heated oxygen sensor for tip damage
			  ▪ Check heated oxygen sensor for tip blocked
			  ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Complete oil consumption check
			  ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel

P2274-00	O2 Sensor Signal Biased/Stuck Lean Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Activity of the sensor is too low
		▪ Prioritized List of Possible Causes
		  ▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Air leakage between heated oxygen sensor and catalyst
		  ▪ Heated oxygen sensor tip damaged
		  ▪ Heated oxygen sensor tip blocked
		  ▪ Heated oxygen sensor tip contaminated by excessive oil consumption
		  ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel
			NOTE:
			This DTC may set as a result of P0191-85 setting. Complete the repair actions for P0191-85 first, clear the Diagnostic Trouble Code(s) and retest. If this DTC returns, perform the checks below
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Fully warmed-up engine, drive off-idle for a minimum of 25 minutes
			▪ Prioritized Checks to Perform
			  ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Check heated oxygen sensor is correctly installed
			  ▪ Check heated oxygen sensor for tip damage
			  ▪ Check heated oxygen sensor for tip blocked
			  ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Complete oil consumption check
			  ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2275-00	O2 Sensor Signal Biased/Stuck Rich Bank 1 Sensor 3 - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Activity of the sensor is too low ▪ Prioritized List of Possible Causes ▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Air leakage between heated oxygen sensor and catalyst ▪ Heated oxygen sensor tip damaged ▪ Heated oxygen sensor tip blocked ▪ Heated oxygen sensor tip contaminated by excessive oil consumption ▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Fully warmed-up engine, drive off-idle for a minimum of 25 minutes ▪ Prioritized Checks to Perform ▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check heated oxygen sensor is correctly installed ▪ Check heated oxygen sensor for tip damage ▪ Check heated oxygen sensor for tip blocked ▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Complete oil consumption check ▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel
P228C-77	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too Low - Commanded position not reachable	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ The high pressure pump signal is below a threshold ▪ Prioritized List of Possible Causes ▪ Fuel leaking to outside of the fuel system ▪ Fuel leaking into the low pressure system	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running greater than 1 minute ▪ Prioritized Checks to Perform ▪ Check for fuel leaks external of the fuel rail ▪ Check high pressure fuel pumps are not leaking from the high pressure system into the low
P228C-85	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too Low - Signal above allowable range	▪ Purpose of the DTC ▪ To detect unacceptably low pressure in the fuel rail ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ The high pressure pump signal is above a threshold ▪ Prioritized List of Possible Causes ▪ Fuel leak ▪ Low pressure fuel system fault ▪ High pressure fuel pump internal failure ▪ High pressure fuel pump timing incorrect due to incorrect camshaft timing	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running for greater than 1 minute ▪ Prioritized Checks to Perform ▪ Check the fuel system for leaks. Rectify as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for low pressure fuel system related Diagnostic Trouble Code(s) and perform the relevant corrective actions ▪ Install a new high pressure fuel pump ▪ Check the integrity of the timing chain and sprockets. Rectify as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P228D-77	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too High - Commanded position not reachable	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ The high pressure pump signal is above a threshold ▪ Prioritized List of Possible Causes ▪ Fuel leaking to outside of the fuel system ▪ Fuel leaking into the low pressure system	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running greater than 1 minute ▪ Prioritized Checks to Perform ▪ Check for fuel leaks external of the fuel rail ▪ Check high pressure fuel pumps are not leaking from the high pressure system into the low
P228D-84	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too High - Signal below allowable range	▪ Purpose of the DTC ▪ To detect unacceptably high pressure in the fuel rail ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ The high pressure pump signal is below a threshold ▪ Prioritized List of Possible Causes ▪ High pressure fuel pump internal failure ▪ High pressure fuel pump timing incorrect due to incorrect camshaft timing	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running for greater than 1 minute ▪ Prioritized Checks to Perform ▪ Install a new high pressure fuel pump ▪ Check the integrity of the timing chain and sprockets. Rectify as necessary
P228E-84	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too Low - Signal below allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ The high pressure pump signal is below a threshold ▪ Prioritized List of Possible Causes ▪ Mechanical failure of high pressure pumps ▪ Worn high pressure pumps ▪ Mis-built engine	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running greater than 1 minute ▪ Prioritized Checks to Perform ▪ Perform check on high pressure pumps operation ▪ Check timing of the fuel pump drive chain

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P228F-85	Fuel Pressure Regulator 1 Exceeded Learning Limits - Too High - Signal above allowable range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ The high pressure pump signal is above a threshold
		▪ Prioritized List of Possible Causes
		  ▪ Mechanical failure of high pressure pumps
		  ▪ Worn high pressure pumps
		  ▪ Mis-built engine
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running greater than 1 minute
			▪ Prioritized Checks to Perform
			  ▪ Perform check on high pressure pumps operation
			  ▪ Check timing of the fuel pump drive chain

P2299-23	Brake Pedal Position/Accelerator Pedal Position Incompatible - Signal stuck low	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module measures a signal that remains low when changes are expected
		  ▪ Brake pedal switch detached from mounting bracket/pedal box while electrically connected
		  ▪ Brake pedal switch mounting position incorrectly adjusted
		  ▪ Brake pedal switch plunger partially stuck in
		  ▪ Brake pedal switch 1 circuit open circuit, short circuit to power, short circuit to ground
		  ▪ Brake pedal switch 2 circuit open circuit, short circuit to power, short circuit to ground
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ With engine running press brake pedal to floor and hold for 1 minute, if fault is present DTC will log
			▪ Prioritized Checks to Perform
			  ▪ Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index
			  ▪ Check that the brake pedal switch is mounted and correctly positioned on the mounting bracket/pedal box
			  ▪ Check for smooth operation of the brake pedal switch plunger
			  ▪ Refer to the electrical circuit diagrams and check brake pedal switch 1 circuit for open circuit, short circuit to power, short circuit to ground
			  ▪ Refer to the electrical circuit diagrams and check brake pedal switch 2 circuit for open circuit, short circuit to power, short circuit to ground

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2299-24	Brake Pedal Position/Accelerator Pedal Position Incompatible - Signal stuck high	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Prioritized List of Possible Causes ▪ The powertrain control module measures a signal that remains high when changes are expected ▪ Brake pedal switch detached from mounting bracket/pedal box while electrically connected ▪ Brake pedal switch mounting position incorrectly adjusted ▪ Brake pedal switch plunger partially stuck out ▪ Brake pedal switch 1 circuit open circuit, short circuit to power, short circuit to ground ▪ Brake pedal switch 2 circuit open circuit, short circuit to power, short circuit to ground	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ With engine running press brake pedal to floor and hold for 1 minute, if fault is present DTC will log ▪ Prioritized Checks to Perform ▪ Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check that the brake pedal switch is mounted and correctly positioned on the mounting bracket/pedal box ▪ Check for smooth operation of the brake pedal switch plunger ▪ Refer to the electrical circuit diagrams and check brake pedal switch 1 circuit for open circuit, short circuit to power, short circuit to ground ▪ Refer to the electrical circuit diagrams and check brake pedal switch 2 circuit for open circuit, short circuit to power, short circuit to ground
P2300-11	Ignition Coil "A" Primary Control Circuit Low - Circuit short to ground	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_P_IOS1 - ▪ Prioritized List of Possible Causes ▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ignition coil circuit, short circuit to ground ▪ Ignition coil failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground ▪ Check and install a new ignition coil only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2301-12	Ignition Coil "A" Primary Control Circuit High - Circuit short to battery	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_P_IOS1 - ▪ Prioritized List of Possible Causes ▪ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ignition coil circuit, short circuit to power ▪ Ignition coil failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to power ▪ Check and install a new ignition coil only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P2303-11	Ignition Coil "B" Primary Control Circuit Low - Circuit short to ground	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_P_IOS2 - ▪ Prioritized List of Possible Causes ▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Ignition coil circuit, short circuit to ground ▪ Ignition coil failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground ▪ Check and install a new ignition coil only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2304-12	Ignition Coil "B" Primary Control Circuit High - Circuit short to battery	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ No
		• Control Module Cavity
		  ◦ Circuit reference - O_P_IOS2 -
		• Prioritized List of Possible Causes
		  ◦ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ Ignition coil circuit, short circuit to power
		  ◦ Ignition coil failure
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to power
			  ◦ Check and install a new ignition coil only when diagnosed as failed
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2306-11	Ignition Coil "C" Primary Control Circuit Low - Circuit short to ground	• Electrical Cause
		  ◦ Yes
		• Mechanical Cause
		  ◦ No
		• Control Module Cavity
		  ◦ Circuit reference - O_P_IOS3 -
		• Prioritized List of Possible Causes
		  ◦ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
		  ◦ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ◦ Ignition coil circuit, short circuit to ground
		  ◦ Ignition coil failure
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ◦ Engine running
			• Prioritized Checks to Perform
			  ◦ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ◦ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground
			  ◦ Check and install a new ignition coil only when diagnosed as failed
			  ◦ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2307-12	Ignition Coil "C" Primary Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS3 -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil circuit, short circuit to power
		  ▪ Ignition coil failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to power
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2309-11	Ignition Coil "D" Primary Control Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS4 -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil circuit, short circuit to ground
		  ▪ Ignition coil failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to ground
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2310-12	Ignition Coil "D" Primary Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_P_IOS4 -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Ignition coil circuit, short circuit to power
		  ▪ Ignition coil failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the ignition coil circuit for short circuit to power
			  ▪ Check and install a new ignition coil only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2400-00	Evaporative Emission System Leak Detection Pump Control Circuit/Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_TDMP -
		▪ Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system leak detection pump power stage driver for correct operation
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Diagnostic module tank leakage pump circuit open circuit, high resistance
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on and engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for open circuit, high resistance

P2401-00	Evaporative Emission System Leak Detection Pump Control Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_TDMP -
		▪ Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system leak detection pump power stage driver for correct operation
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Diagnostic module tank leakage pump circuit short circuit to ground
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on and engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to ground

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2402-00	Evaporative Emission System Leak Detection Pump Control Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_S_TDMP - ▪ Monitor Description ▪ The diagnostic function monitors the evaporative emission system leak detection pump power stage driver for correct operation ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diagnostic module tank leakage pump circuit short circuit to power	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on and engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage pump circuit for short circuit to power
P2404-00	Evaporative Emission System Leak Detection Pump Sense Circuit Range/Performance - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Monitor Description ▪ Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for up to 900 seconds ▪ Prioritized List of Possible Causes ▪ Diagnostic module tank leakage module internal failure	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Make sure that the vehicle has a fuel level between 15 and 85% ▪ Ambient temperature is between 32°F and 95°F ▪ Using the Jaguar Land Rover approved diagnostic equipment complete "large leak test" ▪ Prioritized Checks to Perform ▪ Install a new diagnostic module tank leakage module as necessary
P2405-00	Evaporative Emission System Leak Detection Pump Sense Circuit Low - No sub type information	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for up to 900 seconds ▪ Prioritized List of Possible Causes ▪ Diagnostic module tank leakage module internal failure	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Make sure that the vehicle has a fuel level between 15 and 85% ▪ Ambient temperature is between 32°F and 95°F ▪ Using the Jaguar Land Rover approved diagnostic equipment complete "large leak test" ▪ Prioritized Checks to Perform ▪ Install a new diagnostic module tank leakage module as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2406-00	Evaporative Emission System Leak Detection Pump Sense Circuit High - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for up to 900 seconds
		▪ Prioritized List of Possible Causes
		  ▪ Diagnostic module tank leakage module internal failure
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Make sure that the vehicle has a fuel level between 15 and 85%
			  ▪ Ambient temperature is between 32°F and 95°F
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete "large leak test"
			▪ Prioritized Checks to Perform
			  ▪ Install a new diagnostic module tank leakage module as necessary

P2407-00	Evaporative Emission System Leak Detection Pump Sense Circuit Intermittent/Erratic - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Engine off, diagnostic module tank leakage module drives the internal diagnostic module tank leakage pump and solenoid valve, can operate for up to 900 seconds
		▪ Prioritized List of Possible Causes
		  ▪ Diagnostic module tank leakage module internal failure - Water or fuel vapor detected in the module
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Make sure that the vehicle has a fuel level between 15 and 85%
			  ▪ Ambient temperature is between 32°F and 95°F
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete "large leak test"
			▪ Prioritized Checks to Perform
			  ▪ Inspect the inlet and outlet ports of the module for moisture, If moisture is present then dry out the module, re-fit when dry and re-check
			  ▪ Install a new diagnostic module tank leakage module as necessary

P240A-00	Evaporative Emission System Leak Detection Pump Heater Control Circuit/Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_S_TDMH -
		▪ Monitor Description
		  ▪ The diagnostic function monitors the evaporative emission system leak detection pump heater power stage driver for correct operation
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Diagnostic module tank leakage heater circuit open circuit, high resistance
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on and engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for open circuit, high resistance

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P240B-00	Evaporative Emission System Leak Detection Pump Heater Control Circuit Low - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_S_TDMH - ▪ Monitor Description ▪ The diagnostic function monitors the evaporative emission system leak detection pump heater power stage driver for correct operation ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diagnostic module tank leakage heater circuit short circuit to ground	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on and engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for short circuit to ground
P240C-00	Evaporative Emission System Leak Detection Pump Heater Control Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_S_TDMH - ▪ Monitor Description ▪ The diagnostic function monitors the evaporative emission system leak detection pump heater power stage driver for correct operation ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diagnostic module tank leakage heater circuit short circuit to power	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on and engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage heater circuit for short circuit to power
P2418-00	Evaporative Emission Control System Switching Valve Control Circuit/Open - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - O_S_TDMV - ▪ Monitor Description ▪ The diagnostic function monitors the evaporative emission control system switching valve power stage driver for correct operation ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diagnostic module tank leakage changeover valve circuit open circuit, high resistance	**NOTE:** A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on and engine running ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for open circuit, high resistance

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2419-00	Evaporative Emission Control System Switching Valve Control Circuit Low - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference - O_S_TDMV -
		- Monitor Description
		  - The diagnostic function monitors the evaporative emission control system switching valve power stage driver for correct operation
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Diagnostic module tank leakage changeover valve circuit short circuit to ground
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Ignition on and engine running
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for short circuit to ground

P2420-00	Evaporative Emission Control System Switching Valve Control Circuit High - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference - O_S_TDMV -
		- Monitor Description
		  - The diagnostic function monitors the evaporative emission control system switching valve power stage driver for correct operation
		- Prioritized List of Possible Causes
		  - Connector is disconnected, connector pin is backed out, connector pin corrosion
		  - Diagnostic module tank leakage changeover valve circuit short circuit to power
			NOTE:
			A guided flow is available for this DTC. To assist with correct diagnosis, please make sure to run the guided flow.
			- Vehicle Conditions to Enable DTC Logging Strategy
			  - Ignition on and engine running
			- Prioritized Checks to Perform
			  - Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  - Refer to the electrical circuit diagrams and check the diagnostic module tank leakage changeover valve circuit for short circuit to power

P243F-00	Particulate Filter Restriction - Soot Accumulation Too High Bank 2 - No sub type information	- MIL
		  - No
		- Electrical Cause
		  - No
		- Mechanical Cause
		  - Yes
		- Monitor Description
		  - Particulate Filter Restriction - Soot Accumulation exceeded torque limit threshold
		- Prioritized List of Possible Causes
		- Other Gasoline Particulate Filter (GPF) related Diagnostic Trouble Code(s)
		- Gasoline particulate filter is getting full of soot
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			- Vehicle Conditions to enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			- Check engine control module for GPF related Diagnostic Trouble Code(s) and refer to relevant DTC index
			- Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2452-41	Particulate Filter Pressure Sensor "A" Circuit - General checksum failure	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		▪ Exhaust back pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Gasoline particulate filter pressure sensor failure
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Bank 1 Particulate Filter - Exhaust Back Pressure - 0X0645
			  ▪ With the ignition on, engine not running, check the pressure sensor value, this should be close to 0 (no gas flow)
			  ▪ If the sensor is reading above this then the sensor is offset and requires the following checks before replacement
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check the sensor harness for chaffing or heat damage
			▪ Check gasoline particulate filter pressure sensor hoses for crushed, blockage, splits
			▪ Check and install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

P2452-96	Particulate Filter Pressure Sensor "A" Circuit - Component internal failure	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		▪ Gasoline particulate filter pressure sensor detects the input pressure is too high or too low
		▪ Reverse flowing pressure is detected on the gasoline particulate filter pressure sensor
		▪ Gasoline particulate filter pressure sensor failure
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Bank 1 Particulate Filter - Exhaust Back Pressure - 0X0645
			  ▪ With the ignition on, engine not running, check the pressure sensor value, this should be close to 0 (no gas flow)
			  ▪ If the sensor is reading above this then the sensor is offset and requires the following checks before replacement
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check the sensor harness for chaffing or heat damage
			▪ Check gasoline particulate filter pressure sensor hoses for crushed, blockage, splits
			▪ Check and install a new gasoline particulate filter pressure sensor, only when diagnosed as failed
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2453-28	Particulate Filter Pressure Sensor "A" Circuit Range/Performance - Signal bias level out of range / zero adjustment failure	▪ MIL
		  ▪ No
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Pressure sensor offset is greater than threshold detected during afterrun bank 1
		▪ Prioritized List of Possible Causes
		▪ Differential pressure sensor is offset bank 1
		▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		▪ Differential pressure sensor hoses for crushed, blockage, split
		▪ Differential pressure sensor failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Transition from engine running to afterrun
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Bank 1 Particulate Filter - Exhaust Back Pressure - 0X0645
			  ▪ With the ignition on, engine not running, check the pressure sensor value, this should be close to 0 (no gas flow)
			  ▪ If the sensor is reading above this then the sensor is offset and requires the following checks before replacement
			▪ Refer to the electrical circuit diagrams and check the differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Check the sensor harness for chaffing or heat damage
			▪ Check gasoline particulate filter pressure sensor hoses for crushed, blockage, splits
			▪ Check and install a new gasoline particulate filter pressure sensor, only when diagnosed as failed
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the Diagnostic Trouble Code(s) and retest

P2453-2A	Particulate Filter Pressure Sensor "A" Circuit Range/Performance - Signal stuck in range	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Pressure sensor value not as expected at a given exhaust mass flow Bank 1. A change in exhaust gas flow should have a matching change in exhaust gas pressure. This diagnostic checks for a suitable change in pressure when exhaust gas volume flow is seen
		▪ Prioritized List of Possible Causes
		▪ Differential pressure sensor output value is stuck
		▪ Differential pressure sensor hoses and pipes
		▪ Differential pressure sensor failure
		▪ Gasoline particulate filter mechanical integrity
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running, operate the vehicle through high and low regions of exhaust mass flow
			▪ Prioritized Checks to Perform
			▪ Check differential pressure sensor hoses and pipes for mechanical integrity
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, check datalogger signal - Bank 1 Particulate Filter - Exhaust Back Pressure - 0X0645
			  ▪ With the ignition on, engine not running, check the pressure sensor value, this should be close to 0 (no gas flow)
			  ▪ If the sensor is reading above this then the sensor is offset and requires the following checks before replacement
			▪ Using the Jaguar Land Rover Approved Diagnostic Equipment, clear the Diagnostic Trouble Code(s) and retest. If the fault persists, install a new differential pressure sensor, only when diagnosed as failed
			▪ Check gasoline particulate filter for mechanical integrity, install new, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2453-4A	Particulate Filter Pressure Sensor "A" Circuit Range/Performance - Incorrect component installed	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Wrong sensor ID detected by the powertrain control module bank 1
		▪ Prioritized List of Possible Causes
		▪ Differential pressure sensor incorrect part is installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			▪ Check differential pressure sensor correct part is installed

P2458-00	Particulate Filter Regeneration Duration Bank 1 - No sub type information	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Regeneration demand of particulate filter too high - Self cleaning of particle filter deactivated
		▪ Prioritized List of Possible Causes
		▪ Soot model value reached an exceedingly high value
		▪ Gasoline particulate filter blocked
		▪ Exhaust back pressure sensor hoses split
		▪ Exhaust system leak upstream of the gasoline particulate filter
		▪ Exhaust back pressure sensor failure
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals
			  ▪ Exhaust Particulate Filter Soot Mass (0x061F)
			▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure
			  ▪ Note: If regeneration fails to complete on first attempt, restart the engine, check if the warning message is still present, and restart the service regeneration
			▪ Advise the customer of the driving routine required to regenerate gasoline particulate filter as stated in the owner handbook
			▪ Check the exhaust system for contamination. Rectify as necessary
			▪ Check the exhaust back pressure sensor hoses for blockages and splits. Rectify as necessary. Install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2463-00	Particulate Filter Restriction - Soot Accumulation Bank 1 - No sub type information	▪ MIL
		  ▪ No
		▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Regeneration demand of gasoline particulate filter is higher than second threshold (red message) - Driving cycle not suitable to successfully clean the gasoline particulate filter after a prolonged period of time trying. Restricted performance mode may become active in the following driving cycles if no action is taken
		▪ Prioritized List of Possible Causes
		▪ Soot model value reached very high value
		▪ Gasoline particulate filter obstructed or blocked
		▪ Exhaust back pressure sensor hoses split
		▪ Exhaust system leak upstream of the gasoline particulate filter
		▪ Exhaust back pressure sensor failure
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals
			  ▪ Exhaust Particulate Filter Soot Mass (0x061F)
			▪ Check accumulated soot value using the Jaguar Land Rover approved diagnostic equipment and perform relevant corrective actions
			▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure
			  ▪ Note: If regeneration fails to complete on first attempt, restart the engine, check if the warning message is still present, and restart the service regeneration
			▪ Advise the customer of the driving routine required to regenerate gasoline particulate filter as stated in the owner handbook
			▪ Check the exhaust system for contamination. Rectify as necessary
			▪ Check the exhaust back pressure sensor hoses for blockages and splits. Rectify as necessary. Install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P246B-00	Vehicle Conditions Incorrect for Particulate Filter Regeneration - No sub type information	▪ MIL ▪ NO ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Regeneration demand of gasoline particulate filter is higher than lower threshold - Drive cycle not suitable to successfully clean the gasoline particulate filter ▪ Prioritized List of Possible Causes ▪ Soot model value reached a its threshold value ▪ Gasoline particulate filter blocked ▪ Exhaust back pressure sensor hoses split ▪ Exhaust system leak upstream of the gasoline particulate filter ▪ Exhaust back pressure sensor failure	**NOTE:** The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals ▪ Exhaust Particulate Filter Soot Mass (0x061F) ▪ Check accumulated soot value using the Jaguar Land Rover approved diagnostic equipment and perform relevant corrective actions ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure ▪ Advise the customer of the driving routine required to regenerate gasoline particulate filter as stated in the owner handbook ▪ Check the exhaust system for contamination. Rectify as necessary ▪ Check the exhaust back pressure sensor hoses for blockages and splits. Rectify as necessary. Install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P246C-00	Particulate Filter Restriction - Forced Limited Power Bank 1 - No sub type information	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Soot load exceeded limp home mode threshold - torque limitation active on bank 1. Regeneration demand of gasoline particulate filter is higher than second threshold (red warning - Driving cycle not suitable to successfully clean the gasoline particulate filter after several attempts. Self cleaning measures inhibited for component safety. Only Rapid Service Regeneration allowed. Driving on road will not resolve the issue
		▪ Prioritized List of Possible Causes
		▪ Soot model value reached an exceedingly high value
		▪ Gasoline particulate filter blocked
		▪ Exhaust back pressure sensor hoses split
		▪ Exhaust system leak upstream of the gasoline particulate filter
		▪ Exhaust back pressure sensor failure
			NOTE: The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals
			  ▪ Exhaust Particulate Filter Soot Mass (0x061F)
			▪ Check accumulated soot value using the Jaguar Land Rover approved diagnostic equipment and perform relevant corrective actions
			▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure
			  ▪ Note: If regeneration fails to complete on first attempt, restart the engine, check if the warning message is still present, and restart the service regeneration
			▪ Advise the customer of the driving routine required to regenerate gasoline particulate filter as stated in the owner handbook
			▪ Check the exhaust system for contamination. Rectify as necessary
			▪ Check the exhaust back pressure sensor hoses for blockages and splits. Rectify as necessary. Install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P24A4-00	Particulate Filter Restriction - Soot Accumulation Too High Bank 1 - No sub type information	▪ MIL
		  ▪ No
		▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Monitor Description
		  ▪ Back pressure in the gasoline particulate filter above the back pressure limit threshold
		▪ Prioritized List of Possible Causes
		▪ Gasoline particulate filter system fault
		▪ Exhaust system obstructed or blocked
		▪ Exhaust system contaminated
		▪ Exhaust back pressure sensor hoses blocked or split
		▪ Gasoline particulate filter mechanical integrity compromised
		▪ Regeneration entry conditions not achieved. Customer driving routine does not allow the system to clean the gasoline particulate filter
		▪ Exhaust back pressure sensor failure
			NOTE:
			The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals
			  ▪ Exhaust Particulate Filter Soot Mass (0x061F)
			▪ Check accumulated soot value using the Jaguar Land Rover approved diagnostic equipment and perform relevant corrective actions
			▪ Using the Jaguar Land Rover approved diagnostic equipment, complete the "Particulate Filter Regeneration" (static) procedure
			▪ Advise the customer of the driving routine required to regenerate gasoline particulate filter as stated in the owner handbook
			▪ Check the exhaust system for contamination. Rectify as necessary
			▪ Check the exhaust back pressure sensor hoses for blockages and splits. Rectify as necessary. Install a new gasoline particulate filter pressure sensor, only when diagnosed as failed

P250F-00	Engine Oil Level Too Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_V_5VOPTS -
		  ▪ Circuit reference - I_T_OPTS -
		  ▪ Circuit reference - G_R_OPTS -
		▪ Prioritized List of Possible Causes
		  ▪ Engine oil level too low
		  ▪ Oil pressure and temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Oil pressure and temperature sensor failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check for engine oil leaks. Rectify as necessary. Top up the engine oil as necessary
			  ▪ Refer to the electrical circuit diagrams and check the oil pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness or install a new oil pressure and temperature sensor as necessary
			  ▪ Install a new oil pressure and temperature sensor

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2541-00	Low Pressure Fuel System Sensor Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_FLPS -
		  ▪ Circuit reference - O_V_5VFLPS -
		▪ Monitor Description
		  ▪ Low pressure fuel pressure sensor voltage below threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel low pressure sensor circuit, short circuit to ground
		  ▪ Fuel low pressure sensor 5 volt power supply circuit, open circuit, high resistance
		  ▪ Wear of the low pressure fuel pump
		  ▪ Leaking fuel system external to tank
		  ▪ Leaking fuel system internal to the tank
		  ▪ Fuel low pressure sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on engine running for greater than 2 minutes
			▪ Prioritized Checks to Perform
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground
			  ▪ Check fuel low pressure sensor 5 volt power supply circuit for open circuit, high resistance
			  ▪ Confirm no fuel leaks on vehicle
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new fuel low pressure sensor as required

P2542-00	Low Pressure Fuel System Sensor Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_FLPS -
		  ▪ Circuit reference - G_R_FLPS -
		▪ Monitor Description
		  ▪ Low pressure fuel pressure sensor voltage above threshold
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Fuel low pressure sensor circuit, short circuit to power
		  ▪ Fuel low pressure sensor ground supply circuit, open circuit, high resistance
		  ▪ Leaking fuel system external to tank
		  ▪ Leaking fuel system internal to the tank
		  ▪ Pressure regulator blocked inside tank
		  ▪ Fuel low pressure sensor failure
			▪ Vehicle Conditions to enable DTC Logging strategy
			  ▪ Ignition on engine running for greater than 2 minutes
			▪ Prioritized Checks to Perform
			  ▪ Check connector is not disconnected, connector pin is not backed out
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to power
			  ▪ Check fuel low pressure sensor ground supply circuit for open circuit, high resistance
			  ▪ Confirm no fuel leaks on vehicle
			  ▪ Check pressure regulator inside tank is not blocked
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
			  ▪ Check and install a new fuel low pressure sensor as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P25A9-13	Piston Cooling Oil Control Circuit/Open - Circuit open	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_S_OSJS - ▪ Monitor Description ▪ Open load condition detected on the PCM pin output to piston cooling oil valve ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Piston cooling oil jets solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Piston cooling oil jets solenoid mechanical integrity ▪ Piston cooling oil jets solenoid failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the piston cooling oil jets solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check and install a new piston cooling oil jets solenoid only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P25AA-11	Piston Cooling Oil Control Circuit Low - Circuit short to ground	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Signal Name ▪ Circuit reference - O_S_OSJS - ▪ Monitor Description ▪ Short circuit to ground condition detected on the PCM pin output to piston cooling oil valve ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Piston cooling oil jets solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Piston cooling oil jets solenoid mechanical integrity ▪ Piston cooling oil jets solenoid failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the piston cooling oil jets solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check and install a new piston cooling oil jets solenoid only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P25AB-12	Piston Cooling Oil Control Circuit High - Circuit short to battery	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Signal Name
		  ▪ Circuit reference - O_S_OSJS -
		▪ Monitor Description
		  ▪ Short circuit to battery condition detected on the PCM pin output to piston cooling oil valve
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Piston cooling oil jets solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Piston cooling oil jets solenoid mechanical integrity
		  ▪ Piston cooling oil jets solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the piston cooling oil jets solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new piston cooling oil jets solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P25AC-4B	Piston Cooling Oil Control Circuit Performance / Stuck Off - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Signal Name
		  ▪ Circuit reference - O_S_OSJS -
		▪ Monitor Description
		  ▪ Over temperature condition detected on the PCM internal circuit driving piston cooling oil valve
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Piston cooling oil jets solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Piston cooling oil jets solenoid mechanical integrity
		  ▪ Piston cooling oil jets solenoid failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the piston cooling oil jets solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new piston cooling oil jets solenoid only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P25B3-00	Turbocharger/Supercharger Wastegate "A" Stuck Open - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_SCPOS - ▪ Circuit reference - O_T_SCNEG - ▪ Prioritized List of Possible Causes ▪ Turbocharger electronic wastegate mechanical integrity ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Turbocharger electronic wastegate failure	**NOTE:** If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod ▪ Vehicle Conditions to Enable DTC Setting Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check turbocharger electronic wastegate for mechanical integrity. Rectify as necessary ▪ Check turbocharger wastegate rod/lever for any mechanical damage/bend For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing). ▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation ▪ If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod ▪ Check the voltage at wastegate closed position. If incorrect learned values are seen, clear the learned values and run the learning procedure again ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Install a new turbocharger ▪ Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P25B4-00	Turbocharger/Supercharger Wastegate "A" Stuck Closed - No sub type information	- Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference - O_T_SCPOS - - Circuit reference - O_T_SCNEG - - Prioritized List of Possible Causes - Turbocharger electronic wastegate mechanical integrity - Connector is disconnected, connector pin is backed out, connector pin corrosion - Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance - Turbocharger electronic wastegate failure	**NOTE:** If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod - Vehicle Conditions to Enable DTC Setting Strategy - Engine running - Prioritized Checks to Perform - Check turbocharger electronic wastegate for mechanical integrity. Rectify as necessary - Check turbocharger wastegate rod/lever for any mechanical damage/bend For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing). - Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation - If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod - Check the voltage at wastegate closed position. If incorrect learned values are seen, clear the learned values and run the learning procedure again - Inspect connectors for signs of water ingress and pins for damage and/or corrosion - Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Install a new turbocharger - Using the Jaguar Land Rover approved diagnostic equipment, run application - ECU Diagnostics / Powertrain Control Module (PCM) / Clear Diagnostic Trouble Code(s). Retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2600-13	Coolant Pump "A" Control Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CWP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Auxiliary coolant pump fuse failure
		  ▪ Auxiliary coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Auxiliary coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the auxiliary coolant pump fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new auxiliary coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2601-4B	Coolant Pump "A" Control Circuit Performance / Stuck Off - Over temperature	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CWP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Auxiliary coolant pump fuse failure
		  ▪ Auxiliary coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Auxiliary coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the auxiliary coolant pump fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new auxiliary coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2602-11	Coolant Pump "A" Control Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CWP -
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Auxiliary coolant pump fuse failure
		  ▪ Auxiliary coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Auxiliary coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the auxiliary coolant pump fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new auxiliary coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2603-12	Coolant Pump "A" Control Circuit High - Circuit short to battery	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_CWP - ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Auxiliary coolant pump fuse failure ▪ Auxiliary coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Auxiliary coolant pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check the integrity of the auxiliary coolant pump fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the auxiliary coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check and install a new auxiliary coolant pump only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
P2610-00	ECM/PCM Engine Off Timer Performance - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ The powertrain control module monitors the global time and checks if it plausible ▪ Prioritized List of Possible Causes ▪ Body control module power or ground circuit open circuit, high resistance ▪ Communication bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Body system fault	**NOTE:** The powertrain control module is unlikely to be the root cause for this DTC and must not be replaced without a Technical Assist Request (TA) being raised for further guidance ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, Engine running ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the body control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the communication bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P2610-84	ECM/PCM Engine Off Timer Performance - Signal below allowable range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ The powertrain control module will set a DTC if the engine off time calculated from the global time in the instrument cluster/body control module is less than 1hr, when the shutdown temperature was fully warm and the coolant 1 temperature at start is significantly cooled down to ambient temp ▪ Prioritized List of Possible Causes ▪ Vehicle battery has been isolated and the global time does not increment. Warm coolant has been replaced with cold coolant ▪ Rapid engine temperature cooling with rapid ambient temperature rise	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity for coolant temperature sensor and ambient temperature sensor ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest
P2610-87	ECM/PCM Engine Off Timer Performance - Missing message	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Powertrain control module monitors the global time. If global time is not received then this DTC is set ▪ Prioritized List of Possible Causes ▪ Body control module power or ground circuit open circuit, high resistance ▪ Communication bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Body system fault	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, Engine running ▪ Prioritized Checks to Perform ▪ Check powertrain control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a communication network integrity test ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
P2635-7B	Fuel Pump "A" Low Flow / Performance - Low fluid level	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_PSP - ▪ Prioritized List of Possible Causes ▪ Fuel level too low ▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank ▪ Fuel pump driver module failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Check sufficient fuel is available ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed ▪ Check and install a new fuel pump driver module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2635-92	Fuel Pump "A" Low Flow / Performance - Performance or incorrect operation	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_PSP -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the component performance is outside its expected range or operating in an incorrect way
		  ▪ Fuel level too low
		  ▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank
		  ▪ Fuel pump driver module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check sufficient fuel is available
			  ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed
			  ▪ Check and install a new fuel pump driver module as required

P2635-97	Fuel Pump "A" Low Flow / Performance - Component or system operation obstructed or blocked	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_PSP -
		▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module has detected that the operation of a component is prevented by an obstruction
		  ▪ Fuel level too low
		  ▪ Fuel low pressure sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Fuel pick up pipe is disconnected from the low pressure fuel pump within the fuel tank
		  ▪ Fuel pump driver module failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Check sufficient fuel is available
			  ▪ Refer to electrical circuit diagrams and check the fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
			  ▪ Refer to the relevant sections of the workshop manual and check the fuel system pipework is correctly installed
			  ▪ Check and install a new fuel pump driver module as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P26CA-13	Engine Coolant Pump Control Circuit/Open - Circuit open	▪ Variable coolant pump solenoid connector loose, damaged or corroded
		▪ Water ingress to variable coolant pump solenoid connector
			▪ Inspect the variable coolant pump solenoid connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P26CA-4B	Engine Coolant Pump Control Circuit/Open - Over temperature	▪ Information only. This DTC can be ignored	▪ Information only. This DTC can be ignored
P26CB-72	Engine Coolant Pump Flow Control Valve Stuck Closed - Actuator stuck open	▪ Variable coolant pump slow to respond due to high seal friction – **Do not replace the coolant pump** ▪ Excessive shroud friction due to internal lip seal geometry no longer adhering to specifications ▪ Variable coolant pump position sensor fault ▪ Variable coolant pump control solenoid sensor fault	**NOTES:** ▪ **The variable coolant pump should NOT be replaced if this DTC is set** ▪ This Diagnostic Trouble Codes (DTC) is not a cause of Low Coolant level reporting on the Instrument Panel Cluster Control Module (IPC). If the concern is due to "Low Coolant reporting on the IPC" refer to 303-03B - Engine Cooling - Diagnosis and Testing. ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and install latest relevant level of software to the powertrain control module ▪ Clear Diagnostic Trouble Code(s). Complete a standard road test and retest to confirm if the issue is resolved. If the DTC does not reoccur, return the vehicle to the customer ▪ If the DTC re-occurs but vehicle does not overheat, return vehicle to the customer ▪ If vehicle overheats then, using the Jaguar Land Rover approved diagnostic equipment, consult and follow the relevant Guided Flow. If further guidance is required, Raise a TA

P26CC-11	Engine Coolant Pump Control Circuit Low - Circuit short to ground	▪ Variable coolant pump solenoid connector loose, damaged or corroded
		▪ Water ingress to variable coolant pump solenoid connector
			▪ Inspect the variable coolant pump solenoid connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P26CD-12	Engine Coolant Pump Control Circuit High - Circuit short to battery	▪ Variable coolant pump solenoid connector loose, damaged or corroded
		▪ Water ingress to variable coolant pump solenoid connector
			▪ Inspect the variable coolant pump solenoid connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P26DB-13	Engine Sound Control "A" Circuit/Open - Circuit open	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_GSQ -
		▪ Monitor Description
		  ▪ Open load condition detected on the PCM pin audio amplifier
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Audio amplifier circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the audio amplifier circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P26DC-11	Engine Sound Control "A" Circuit Low - Circuit short to ground	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_GSQ -
		▪ Monitor Description
		  ▪ Short circuit to ground condition detected on the PCM pin Audio Amplifier
		▪ Prioritized List of Possible Causes
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Audio amplifier circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Refer to the electrical circuit diagrams and check the audio amplifier circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

<table>
<thead>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
</thead>
<tbody>
<tr>
<td>P26DD-12</td>
<td>Engine Sound Control "A" Circuit High - Circuit short to battery</td>
<td>

Electrical Cause
Yes
Mechanical Cause
No
Control Module Cavity
Circuit reference - O_T_GSQ -
Monitor Description
Short circuit to battery condition detected on the PCM pin Audio Amplifier
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Audio amplifier circuit short circuit to ground, short circuit to power, open circuit, high resistance

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<td>

Vehicle Conditions to Enable DTC Logging Strategy
Engine running
Prioritized Checks to Perform
Inspect connectors for signs of water ingress and pins for damage and/or corrosion
Refer to the electrical circuit diagrams and check the audio amplifier circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

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</tr>
<tr>
<td>P2AB8-16</td>
<td>Wastegate Position Sensor "A" Circuit Low - Circuit voltage below threshold</td>
<td>

Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - I_A_WGP -
Circuit reference - O_V_5VWGPT -
Circuit reference - G_R_WGPT -
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
Turbocharger electronic wastegate failure

</td>
<td>
NOTE:
If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. DO NOT remove the actuator or the connecting rod

Vehicle Conditions to Enable DTC Logging Strategy
Ignition On
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Check the powertrain control module connectors for water ingress
Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement DO NOT attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
Check turbocharger wastegate voltage at closed position
Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: Turbocharger (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

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</tr>
</tbody>
</table>

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr><td>P2AB9-17</td><td>Wastegate Position Sensor "A" Circuit High - Circuit voltage above threshold</td><td>Electrical Cause
Yes
Mechanical Cause
Yes
Control Module Cavity
Circuit reference - I_A_WGP -
Circuit reference - O_V_5VWGPT -
Circuit reference - G_R_WGPT -
Prioritized List of Possible Causes
Connector is disconnected, connector pin is backed out, connector pin corrosion
Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
Turbocharger electronic wastegate failure</td><td>NOTE:
If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. DO NOT remove the actuator or the connecting rod
Vehicle Conditions to Enable DTC Logging Strategy
Ignition On
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
Check the powertrain control module connectors for water ingress
Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement DO NOT attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
Check turbocharger wastegate voltage at closed position
Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: Turbocharger (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest</td></tr>
<tr><td>P2B5D-16</td><td>Engine Coolant Flow Control Valve Position Sensor Circuit - Circuit voltage below threshold</td><td>Variable coolant pump mechanical fault
Shroud beyond maximum allowed closed position
Variable coolant pump position sensor fault
Sensor voltage out of range</td><td>Inspect the variable coolant pump connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
Short circuit to ground
Short circuit to power
Open circuit
High resistance
Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest</td></tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2B5D-17	Engine Coolant Flow Control Valve Position Sensor Circuit - Circuit voltage above threshold	▪ Variable coolant pump mechanical fault
		  ▪ Shroud beyond maximum allowed closed position
		▪ Variable coolant pump position sensor fault
		  ▪ Sensor voltage out of range
			▪ Inspect the variable coolant pump connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2B5E-00	Engine Coolant Flow Control Valve Position Sensor Circuit Low - No sub type information	▪ Variable coolant pump mechanical fault
		  ▪ Shroud beyond maximum allowed closed position
		▪ Variable coolant pump position sensor fault
		  ▪ Sensor voltage out of range
			▪ Inspect the variable coolant pump connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2B5F-00	Engine Coolant Flow Control Valve Position Sensor Circuit High - No sub type information	▪ Variable coolant pump mechanical fault
		  ▪ Shroud beyond maximum allowed closed position
		▪ Variable coolant pump position sensor fault
		  ▪ Sensor voltage out of range
			▪ Inspect the variable coolant pump connector for signs of water ingress and pins for damage and/or corrosion and rectify as required
			▪ Refer to the electrical circuit diagrams and check the variable coolant pump circuit(s) for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2B61-73	Engine Coolant Flow Control Valve Stuck Closed - Actuator stuck closed	▪ Variable coolant pump slow to respond due to high seal friction - **Do not replace the coolant pump**
		▪ Excessive shroud friction due to internal lip seal geometry no longer adhering to specifications
		▪ Variable coolant pump position sensor fault
		▪ Variable coolant pump control solenoid sensor fault
			NOTES:
			▪ **The variable coolant pump should NOT be replaced if this DTC is set**
			▪ This DTC is not a cause of Low Coolant level reporting on the IPC. if the concern is due to "Low Coolant level reporting on the IPC" refer to 303-03B - Engine cooling - Diagnosis and Testing
			
			▪ Vehicle Conditions to Enable DTC Setting Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check and install latest relevant level of software to the powertrain control module
			  ▪ Clear Diagnostic Trouble Code(s). Complete a standard road test and retest to confirm if the issue is resolved. If the DTC does not reoccur, return the vehicle to the customer
			  ▪ If the DTC re-occurs but vehicle does not overheat, return vehicle to the customer

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2B81-16	Wastegate Position Sensor "A" Circuit Performance - Circuit voltage below threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_WGP -
		  ▪ Circuit reference - O_V_5VWGPT -
		  ▪ Circuit reference - G_R_WGPT -
		▪ Prioritized List of Possible Causes
		  ▪ Turbocharger electronic wastegate mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position. If incorrect learned values are seen, clear the learned values and run the learning procedure again
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2B81-17	Wastegate Position Sensor "A" Circuit Performance - Circuit voltage above threshold	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_A_WGP -
		  ▪ Circuit reference - O_V_5VWGPT -
		  ▪ Circuit reference - G_R_WGPT -
		▪ Prioritized List of Possible Causes
		  ▪ Turbocharger electronic wastegate mechanical integrity
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Turbocharger electronic wastegate circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Turbocharger electronic wastegate failure
			NOTE:
			If a fault is identified with the wastegate system or the actuator, install a new turbocharger. The actuator is not a replaceable part. **DO NOT** remove the actuator or the connecting rod
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition On
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Check the powertrain control module connectors for water ingress
			▪ Refer to the electrical circuit diagrams and check the turbocharger electronic wastegate circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check turbocharger wastegate rod movement, start the engine and monitor wastegate rod movement **DO NOT** attempt to disconnect, manipulate or put pressure on the waste gate rod, doing so can result in damage and loss of function of waste gate actuation
			▪ Check turbocharger wastegate voltage at closed position. If incorrect learned values are seen, clear the learned values and run the learning procedure again
			▪ Check turbocharger wastegate rod/lever for any mechanical damage, bend or heavy corrosion close to pin For additional information, refer to: **Turbocharger** (303-04D Fuel Charging and Controls - Turbocharger - INGENIUM I4 2.0L Petrol, Diagnosis and Testing).
			▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the Diagnostic Trouble Code(s) and retest

P2B83-00	Charge Air Cooler Coolant Pump "A" Overspeed/Air In System - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ Yes
		▪ Prioritized List of Possible Causes
		  ▪ Coolant level low
		  ▪ Air trapped in cooling system
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the relevant section of the workshop manual and check that the coolant level is correct, air is not trapped in the cooling system, charge air cooler coolant pump

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P2B84-00	Charge Air Cooler Coolant Pump "A" Underspeed - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CCP -
		▪ Prioritized List of Possible Causes
		  ▪ Coolant system contamination/debris
		  ▪ Charge air cooler inlet hose blocked
		  ▪ Charge air cooler outlet hose blocked
		  ▪ Charge air cooler coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the relevant section of the workshop manual and check the coolant system for contamination/debris
			  ▪ Refer to the relevant section of the workshop manual and check the charge air cooler inlet hose for blockages
			  ▪ Refer to the relevant section of the workshop manual and check the charge air cooler outlet hose for blockages
			  ▪ Check and install a new charge air cooler coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P2BC2-1C	Charge Air Cooler Coolant Pump "A" Supply Voltage Circuit - Circuit voltage out of range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - O_T_CCP -
		▪ Prioritized List of Possible Causes
		  ▪ Coolant system contamination/debris
		  ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		  ▪ Charge air cooler coolant pump fuse failure
		  ▪ Charge air cooler coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Charge air cooler coolant pump failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			  ▪ Refer to the relevant section of the workshop manual and check the coolant system for contamination/debris
			  ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion
			  ▪ Check the integrity of the charge air cooler coolant pump fuse. Rectify as necessary
			  ▪ Refer to the electrical circuit diagrams and check the charge air cooler coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Check and install a new charge air cooler coolant pump only when diagnosed as failed
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

P3400-08	Cylinder Deactivation System Bank 1 - Bus signal/message failures	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ UniAir SV driver module SPI communication failure
		▪ Prioritized List of Possible Causes
		▪ Powertrain control module software issue
		▪ Powertrain control module internal failure
		▪ Battery voltage is low
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check and install latest relevant level of software to the powertrain control module
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3401-00	Cylinder 1 Deactivation/Intake Valve Control Circuit/Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3402-1C	Cylinder 1 Deactivation/Intake Valve Control Circuit Performance - Circuit voltage out of range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3402-7C	Cylinder 1 Deactivation/Intake Valve Control Circuit Performance - Slow response	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH1 ▪ Circuit reference O_T_CVLL1 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3403-00	Cylinder 1 Deactivation/Intake Valve Control Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3403-21	Cylinder 1 Deactivation/Intake Valve Control Circuit Low - Signal amplitude < minimum	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference - G_G_BAT1 -
		  - Circuit reference - G_G_BAT2 -
		  - Circuit reference - G_G_BAT3 -
		  - Circuit reference - V_V_BAT3 -
		  - Circuit reference - V_V_BAT1 -
		  - Circuit reference - V_V_BAT2 -
		- DTC used to determine if the ASIC (Application Specific Integrated Circuit) voltage is to low
		- Prioritized List of Possible Causes
		- Continuously variable valve lift solenoid - Wiring integrity
		  - Short circuit to ground
		  - Short circuit to power
		  - Open circuit
		  - High resistance
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Powertrain control module internal failure
		- Vehicle battery or charging system fault
			- Vehicle Conditions to enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			- Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			- Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  - Short circuit to ground
			  - Short circuit to power
			  - Open circuit
			  - High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			- If connector and/or circuit faults are identified, repair or replace the affected components as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3404-00	Cylinder 1 Deactivation/Intake Valve Control Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH1 ▪ Circuit reference O_T_CVLL1 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3409-00	Cylinder 2 Deactivation/Intake Valve Control Circuit/Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3410-1C	Cylinder 2 Deactivation/Intake Valve Control Circuit Performance - Circuit voltage out of range	- Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference O_T_CVLH2 - Circuit reference O_T_CVLL2 - Prioritized List of Possible Causes - Connector is disconnected, connector pin is backed out, connector pin corrosion - Powertrain control module internal failure - Continuously variable valve lift solenoid - Wiring integrity - Short circuit to ground - Short circuit to power - Open circuit - High resistance - Continuously variable valve lift solenoid failure	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritized Checks to Perform - Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index - Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion - Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for - Short circuit to ground - Short circuit to power - Open circuit - High resistance - wiring resistance should be in a range of 100 mOhm +-10% - If connector and/or circuit faults are identified, repair or replace the affected components as required - Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3410-7C	Cylinder 2 Deactivation/Intake Valve Control Circuit Performance - Slow response	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference O_T_CVLH2
		  - Circuit reference O_T_CVLL2
		- Prioritized List of Possible Causes
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Powertrain control module internal failure
		- Continuously variable valve lift solenoid - Wiring integrity
		  - Short circuit to ground
		  - Short circuit to power
		  - Open circuit
		  - High resistance
		- Continuously variable valve lift solenoid failure
			- Vehicle Conditions to enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			- Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			- Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  - Short circuit to ground
			  - Short circuit to power
			  - Open circuit
			  - High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			- If connector and/or circuit faults are identified, repair or replace the affected components as required
			- Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3411-00	Cylinder 2 Deactivation/Intake Valve Control Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3411-21	Cylinder 2 Deactivation/Intake Valve Control Circuit Low - Signal amplitude < minimum	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference - G_G_BAT1 -
		  ▪ Circuit reference - G_G_BAT2 -
		  ▪ Circuit reference - G_G_BAT3 -
		  ▪ Circuit reference - V_V_BAT3 -
		  ▪ Circuit reference - V_V_BAT1 -
		  ▪ Circuit reference - V_V_BAT2 -
		▪ Purpose of the DTC
		▪ DTC used to determine if the ASIC (Application Specific Integrated Circuit) voltage is to low
		▪ Prioritized List of Possible Causes
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Vehicle battery or charging system fault
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3412-00	Cylinder 2 Deactivation/Intake Valve Control Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3417-00	Cylinder 3 Deactivation/Intake Valve Control Circuit/Open - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3418-1C	Cylinder 3 Deactivation/Intake Valve Control Circuit Performance - Circuit voltage out of range	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3418-7C	Cylinder 3 Deactivation/Intake Valve Control Circuit Performance - Slow response	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3419-00	Cylinder 3 Deactivation/Intake Valve Control Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3419-21	Cylinder 3 Deactivation/Intake Valve Control Circuit Low - Signal amplitude < minimum	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - G_G_BAT1 - ▪ Circuit reference - G_G_BAT2 - ▪ Circuit reference - G_G_BAT3 - ▪ Circuit reference - V_V_BAT3 - ▪ Circuit reference - V_V_BAT1 - ▪ Circuit reference - V_V_BAT2 - ▪ Purpose of the DTC ▪ DTC used to determine if the ASIC (Application Specific Integrated Circuit) voltage is to low ▪ Prioritized List of Possible Causes ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Vehicle battery or charging system fault	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3420-00	Cylinder 3 Deactivation/Intake Valve Control Circuit High - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH3 ▪ Circuit reference O_T_CVLL3 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3425-00	Cylinder 4 Deactivation/Intake Valve Control Circuit/Open - No sub type information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH4 ▪ Circuit reference O_T_CVLL4 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3426-1C	Cylinder 4 Deactivation/Intake Valve Control Circuit Performance - Circuit voltage out of range	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH4 ▪ Circuit reference O_T_CVLL4 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10% ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3426-7C	Cylinder 4 Deactivation/Intake Valve Control Circuit Performance - Slow response	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH4
		  ▪ Circuit reference O_T_CVLL4
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3427-00	Cylinder 4 Deactivation/Intake Valve Control Circuit Low - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH4
		  ▪ Circuit reference O_T_CVLL4
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

<table>
<tr><th>DTC</th><th>DESCRIPTION</th><th>POSSIBLE CAUSES</th><th>ACTION</th></tr>
<tr>
<td>P3427-21</td>
<td>Cylinder 4 Deactivation/Intake Valve Control Circuit Low - Signal amplitude < minimum</td>
<td>
Electrical Cause
Yes
Mechanical Cause
No
Control Module Cavity
Circuit reference - G_G_BAT1 -
Circuit reference - G_G_BAT2 -
Circuit reference - G_G_BAT3 -
Circuit reference - V_V_BAT3 -
Circuit reference - V_V_BAT1 -
Circuit reference - V_V_BAT2 -

Purpose of the DTC
DTC used to determine if the ASIC (Application Specific Integrated Circuit) voltage is to low
Continuously variable valve lift solenoid - Wiring integrity
Short circuit to ground
Short circuit to power
Open circuit
High resistance

Connector is disconnected, connector pin is backed out, connector pin corrosion
Powertrain control module internal failure
Charging system fault
</td>
<td>
Vehicle Conditions to enable DTC Logging Strategy
Engine running
Prioritized Checks to Perform
Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
Short circuit to ground
Short circuit to power
Open circuit
High resistance - wiring resistance should be in a range of 100 mOhm +-10%

If connector and/or circuit faults are identified, repair or replace the affected components as required
</td>
</tr>
</table>

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3428-00	Cylinder 4 Deactivation/Intake Valve Control Circuit High - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH4
		  ▪ Circuit reference O_T_CVLL4
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P3498-00	Cylinder 1 Deactivation Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3498-64	Cylinder 1 Deactivation Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P3498-71	Cylinder 1 Deactivation Performance - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift and/or misfire related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3498-86	Cylinder 1 Deactivation Performance - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH1
		  ▪ Circuit reference O_T_CVLL1
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3498-94	Cylinder 1 Deactivation Performance - Unexpected operation	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		▪ Software failure mechanical angle violation or electrical angle violation (change of requested lift) or device driver programing. Application-specific integrated circuit (ASIC) initialization failure
		▪ Prioritized List of Possible Causes
		▪ Powertrain control module software failure
		▪ Powertrain control module internal failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running in catalyst heating, after a minimum soak period of 2 hours
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			▪ Clear DTCs and retest. If the fault persists, install a new powertrain control module, only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
			  ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors
			  ▪ Make sure PCM side connectors are always dry
			  ▪ Make sure mating PCM harness connectors are always dry
			  ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM
			  ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3499-00	Cylinder 2 Deactivation Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P3499-64	Cylinder 2 Deactivation Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P3499-71	Cylinder 2 Deactivation Performance - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift and/or misfire related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P3499-86	Cylinder 2 Deactivation Performance - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH2
		  ▪ Circuit reference O_T_CVLL2
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P3499-94	Cylinder 2 Deactivation Performance - Unexpected operation	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Software failure mechanical angle violation or electrical angle violation (change of requested lift) or device driver programing. Application-specific integrated circuit (ASIC) initialization failure ▪ Prioritized List of Possible Causes ▪ Powertrain control module software failure ▪ Powertrain control module internal failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running in catalyst heating, after a minimum soak period of 2 hours ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module ▪ Clear DTCs and retest. If the fault persists, install a new powertrain control module, only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors ▪ Make sure PCM side connectors are always dry ▪ Make sure mating PCM harness connectors are always dry ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P349A-00	Cylinder 3 Deactivation Performance - No sub type information	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P349A-64	Cylinder 3 Deactivation Performance - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P349A-71	Cylinder 3 Deactivation Performance - Actuator stuck	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift and/or misfire related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P349A-86	Cylinder 3 Deactivation Performance - Signal invalid	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference O_T_CVLH3
		  ▪ Circuit reference O_T_CVLL3
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Powertrain control module internal failure
		▪ Continuously variable valve lift solenoid - Wiring integrity
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Continuously variable valve lift solenoid failure
		▪ Excessive fuel in engine oil
		▪ Oil pressure is too low
		▪ Oil aeration
		▪ Incorrect specification engine oil installed
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			▪ Install new engine oil to correct specification
			▪ If connector and/or circuit faults are identified, repair or replace the affected components as required
			▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P349A-94	Cylinder 3 Deactivation Performance - Unexpected operation	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		▪ Software failure mechanical angle violation or electrical angle violation (change of requested lift) or device driver programing. Application-specific integrated circuit (ASIC) initialization failure
		▪ Prioritized List of Possible Causes
		▪ Powertrain control module software failure
		▪ Powertrain control module internal failure
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Engine running in catalyst heating, after a minimum soak period of 2 hours
			▪ Prioritized Checks to Perform
			▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module
			▪ Clear DTCs and retest. If the fault persists, install a new powertrain control module, only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module
			  ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors
			  ▪ Make sure PCM side connectors are always dry
			  ▪ Make sure mating PCM harness connectors are always dry
			  ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM
			  ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

P349B-00	Cylinder 4 Deactivation Performance - No sub type information	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - No
		- Control Module Cavity
		  - Circuit reference O_T_CVLH4
		  - Circuit reference O_T_CVLL4
		- Prioritized List of Possible Causes
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Powertrain control module internal failure
		- Continuously variable valve lift solenoid - Wiring integrity
		  - Short circuit to ground
		  - Short circuit to power
		  - Open circuit
		  - High resistance
		- Continuously variable valve lift solenoid failure
			- Vehicle Conditions to enable DTC Logging Strategy
			  - Engine running
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			- Inspect the continuously variable valve lift solenoid circuit connectors for signs of water ingress, pins damage and/or corrosion
			- Refer to the electrical circuit diagrams and check the continuously variable valve lift solenoid circuit for
			  - Short circuit to ground
			  - Short circuit to power
			  - Open circuit
			  - High resistance - wiring resistance should be in a range of 100 mOhm +-10%
			- If connector and/or circuit faults are identified, repair or replace the affected components as required
			- Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

P349B-64	Cylinder 4 Deactivation Performance - Signal plausibility failure	- Electrical Cause
		  - Yes
		- Mechanical Cause
		  - Yes
		- Control Module Cavity
		  - Circuit reference O_T_CVLH4
		  - Circuit reference O_T_CVLL4
		- Prioritized List of Possible Causes
		- Connector is disconnected, connector pin is backed out, connector pin corrosion
		- Powertrain control module internal failure
		- Continuously variable valve lift solenoid - Wiring integrity
		  - Short circuit to ground
		  - Short circuit to power
		  - Open circuit
		  - High resistance
		- Continuously variable valve lift solenoid failure
		- Excessive fuel in engine oil
		- Oil pressure is too low
		- Oil aeration
		- Incorrect specification engine oil installed
			- Vehicle Conditions to enable DTC Logging Strategy
			  - Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C
			- Prioritized Checks to Perform
			- Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index
			- Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index
			- Refer to the relevant sections of the workshop manual and check the oil pressure is within specification
			- Install new engine oil to correct specification
			- If connector and/or circuit faults are identified, repair or replace the affected components as required
			- Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P349B-71	Cylinder 4 Deactivation Performance - Actuator stuck	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH4 ▪ Circuit reference O_T_CVLL4 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure ▪ Excessive fuel in engine oil ▪ Oil pressure is too low ▪ Oil aeration ▪ Incorrect specification engine oil installed	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift and/or misfire related DTCs and refer to the relevant DTC index ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification ▪ Install new engine oil to correct specification ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).
P349B-86	Cylinder 4 Deactivation Performance - Signal invalid	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference O_T_CVLH4 ▪ Circuit reference O_T_CVLL4 ▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Powertrain control module internal failure ▪ Continuously variable valve lift solenoid - Wiring integrity ▪ Short circuit to ground ▪ Short circuit to power ▪ Open circuit ▪ High resistance ▪ Continuously variable valve lift solenoid failure ▪ Excessive fuel in engine oil ▪ Oil pressure is too low ▪ Oil aeration ▪ Incorrect specification engine oil installed	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running for approx 60 seconds when CVVL oil temperature greater than 20°C ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index ▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification ▪ Install new engine oil to correct specification ▪ If connector and/or circuit faults are identified, repair or replace the affected components as required ▪ Check the resistance at the continuously variable valve lift solenoid (resistance should be in a range of 288 mOhm +-10%). If resistance values are not to specification, install a new continuously variable valve lift solenoid following the appropriate service procedures For additional information, refer to: **Continuous Variable Valve Lift** (303-01C Engine - INGENIUM I4 2.0L Petrol, Removal and Installation).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P349B-94	Cylinder 4 Deactivation Performance - Unexpected operation	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Monitor Description ▪ Software failure mechanical angle violation or electrical angle violation (change of requested lift) or device driver programing. Application-specific integrated circuit (ASIC) initialization failure ▪ Prioritized List of Possible Causes ▪ Powertrain control module software failure ▪ Powertrain control module internal failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running in catalyst heating, after a minimum soak period of 2 hours ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check for PCM power supply and/or charging system related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check powertrain control module for continuously variable valve lift related DTCs and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the powertrain control module ▪ Clear DTCs and retest. If the fault persists, install a new powertrain control module, only when diagnosed as failed. Using the Jaguar Land Rover approved diagnostic equipment, configure the powertrain control module as a new module ▪ The powertrain control module now has a liquid cooling system incorporated. The following care points must be followed during removal, installation of component and disconnection, connection of electrical harness connectors ▪ Make sure PCM side connectors are always dry ▪ Make sure mating PCM harness connectors are always dry ▪ Make sure PCM is always handled with connector facing upwards to avoid any accidental exposure of the connector to the coolant from the cooling pipe of the PCM to avoid damage to the PCM ▪ Make sure that the connector from the PCM (harness side) is disconnected and connected only in a dry and clean environment
U0001-88	High Speed CAN Communication Bus - Bus off	▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined failures where a data bus is not available ▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment complete network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0080-82	Vehicle Communication Bus F - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ Communication bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ ABS module fuse failure ▪ HVAC Control Module fuse failure ▪ EPB module fuse failure ▪ Using the Jaguar Land Rover approved diagnostic equipment run routine to read DID 0xD020. The response from this routine will indicate the diagnosis to follow	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the communication bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the ABS module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the HVAC Control Module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the EPB module ▪ Using the Jaguar Land Rover approved diagnostic equipment check for communication network Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment check for Diagnostic Trouble Code(s) in specified control modules (seen in the response from DID 0xD020 request)
U0080-83	Vehicle Communication Bus F - Value of signal protection calculation incorrect	▪ Prioritized List of Possible Causes ▪ Communication bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ ABS module fuse failure ▪ HVAC Control Module fuse failure ▪ EPB module fuse failure ▪ Using the Jaguar Land Rover approved diagnostic equipment run routine to read DID 0xD020. The response from this routine will indicate the diagnosis to follow	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the communication bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the ABS module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the HVAC Control Module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the EPB module ▪ Using the Jaguar Land Rover approved diagnostic equipment check for communication network Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment check for Diagnostic Trouble Code(s) in specified control modules (seen in the response from DID 0xD020 request)

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0080-87	Vehicle Communication Bus F - Missing message	▪ Prioritized List of Possible Causes ▪ Communication bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ ABS module fuse failure ▪ HVAC Control Module fuse failure ▪ EPB module fuse failure ▪ Using the Jaguar Land Rover approved diagnostic equipment run routine to read DID 0xD020. The response from this routine will indicate the diagnosis to follow	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the communication bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the ABS module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the HVAC Control Module ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the EPB module ▪ Using the Jaguar Land Rover approved diagnostic equipment check for communication network Diagnostic Trouble Code(s) ▪ Using the Jaguar Land Rover approved diagnostic equipment check for Diagnostic Trouble Code(s) in specified control modules (seen in the response from DID 0xD020 request)
U0080-88	Vehicle Communication Bus F - Bus off	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference - B_D_FR_BP - ▪ Circuit reference - B_D_FR_BM - ▪ Prioritized List of Possible Causes ▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Anti-lock brake system control module power or ground circuit open circuit, high resistance ▪ Gateway module power or ground circuit open circuit, high resistance	**NOTE:** The FlexRay wiring harness must not be repaired by making a localized wiring repair. A new wiring harness should be installed. ▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on ▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the gateway module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0101-00	Lost Communication With TCM - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and transmission control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the transmission control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the transmission control module ▪ Check the CAN circuit between powertrain control module and transmission control module. Repair as necessary
U0103-00	Lost Communication With Gear Shift Control Module A - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and gear shift module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the gear shift module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check gear shift module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the gear shift module ▪ Check the CAN circuit between powertrain control module and gear shift module. Repair as necessary
U0104-00	Lost Communication With Cruise Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and cruise control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the cruise control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check cruise control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the cruise control module ▪ Check the CAN circuit between powertrain control module and cruise control module. Repair as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U011D-00	Lost Communication With All Wheel Drive Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and all wheel drive control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the all wheel drive control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check all wheel drive control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the all wheel drive control module ▪ Check the CAN circuit between powertrain control module and all wheel drive control module. Repair as necessary
U0121-00	Lost Communication With Anti-Lock Brake System (ABS) Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and anti-lock brake system control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module ▪ Check the CAN circuit between powertrain control module and anti-lock brake system control module. Repair as necessary
U0126-00	Lost Communication With Steering Angle Sensor Module - No sub type information	▪ Prioritized List of Possible Causes ▪ The powertrain control module has not received the expected CAN signal from the steering angle sensor control module within the specified time interval ▪ CAN harness link between powertrain control module and steering angle sensor control module network malfunction	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check steering angle sensor control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check steering angle sensor control module power and ground circuits for open circuit ▪ Check the CAN circuit between powertrain control module and steering angle sensor control module, repair as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0128-00	Lost Communication With Park Brake Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ The powertrain control module has not received the expected CAN signal from the parking brake control module within the specified time interval ▪ CAN harness link between powertrain control module and parking brake control module network malfunction	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check parking brake control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check parking brake control module power and ground circuits for open circuit ▪ Check the CAN circuit between powertrain control module and parking brake control module, repair as necessary
U0131-00	Lost Communication With Power Steering Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and power steering control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the power steering control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check power steering control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the power steering control module ▪ Check the CAN circuit between powertrain control module and power steering control module. Repair as necessary
U0138-00	Lost Communication With All Terrain Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and terrain optimization module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the terrain optimization module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check terrain optimization module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the terrain optimization module ▪ Check the CAN circuit between powertrain control module and terrain optimization module. Repair as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0140-00	Lost Communication With Body Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and body control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the body control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the body control module ▪ Check the CAN circuit between powertrain control module and body control module. Repair as necessary
U0146-00	Lost Communication With Gateway "A" - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and gateway module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the gateway module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment check gateway module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the gateway module ▪ Check the CAN circuit between powertrain control module and gateway module. Repair as necessary
U0151-00	Lost Communication With Restraints Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ CAN harness link between powertrain control module and restraints control module network malfunction ▪ The powertrain control module has not received the expected CAN signal from the restraints control module within the specified time interval	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ If this DTC is set with U0151-00 and B10A2-32, check for fuse failure, restraints control module power and ground circuits for open circuit. Refer to the electrical circuit diagrams and check CAN harness ▪ If this DTC is set with B10A2-32, or on its own refer to the electrical circuit diagrams and check restraints control module crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the Jaguar Land Rover approved diagnostic equipment, check restraints control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check restraints control module power and ground circuits for open circuit. Check the CAN circuit between powertrain control module and restraints control module, repair as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U0155-00	Lost Communication With Instrument Panel Cluster (IPC) Control Module - No sub type information	• Prioritized List of Possible Causes
		  ▪ CAN harness link between powertrain control module and instrument cluster network malfunction
		  ▪ The powertrain control module has not received the expected CAN signal from the instrument cluster within the specified time interval
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Ignition on, engine running
			• Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check instrument cluster for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a CAN network integrity test
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the instrument cluster
			  ▪ Check the CAN circuit between powertrain control module and instrument cluster. Repair as necessary

U0164-00	Lost Communication With HVAC Control Module - No sub type information	• Prioritized List of Possible Causes
		  ▪ HVAC Control Module power or ground circuit open circuit, high resistance
		  ▪ High speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ High speed CAN bus (power mode zero) circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Climate control system fault
			• Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the HVAC Control Module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (power mode zero) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the HVAC Control Module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

U0284-07	Lost Communication with Active Grille Air Shutter Module "A" - Mechanical failures	• Electrical Cause
		  ▪ No
		• Mechanical Cause
		  ▪ Yes
		• Control Module Cavity
		  ▪ Circuit reference - B_D_LIN -
		• Monitor Description
		  ▪ The actuator detected while moving very low torque as when actuator is mechanically disconnected
		• Prioritized List of Possible Causes
		  ▪ Active grille air shutter mechanical failure
			• Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ When actuator is commanded to move, it internally monitors torque required to move. If too low, fault is set meaning expected friction is not present
			• Prioritized Checks to Perform
			  ▪ Check active grille air shutter for mechanical failure

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0284-87	Lost Communication with Active Grille Air Shutter Module "A" - Missing message	▪ Prioritized List of Possible Causes ▪ The powertrain control module has not received the expected CAN signal from the active grille control module within the specified time interval ▪ Other active grille control module related Diagnostic Trouble Code(s) ▪ CAN harness link between powertrain control module and active grille control module network malfunction ▪ Active grille control module power and ground circuits open circuit	▪ Prioritized Checks to Perform ▪ Check active grille control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check active grille control module power and ground circuits for open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
U0285-07	Lost Communication with Active Grille Air Shutter Module "B" - Mechanical failures	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - B_D_LIN - ▪ Monitor Description ▪ The actuator detected while moving very low torque as when actuator is mechanically disconnected ▪ Prioritized List of Possible Causes ▪ Active grille air shutter mechanical failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ When actuator is commanded to move, it internally monitors torque required to move. If too low, fault is set meaning expected friction is not present ▪ Prioritized Checks to Perform ▪ Check active grille air shutter for mechanical failure
U02A9-87	Lost Communication With Charge Air Cooler Coolant Pump "A" - Missing message	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference - O_T_CCP - ▪ Prioritized List of Possible Causes ▪ Coolant system contamination/debris ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Charge air cooler coolant pump fuse failure ▪ Charge air cooler coolant pump circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Charge air cooler coolant pump failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Engine running ▪ Prioritized Checks to Perform ▪ Refer to the relevant section of the workshop manual and check the coolant system for contamination/debris ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Check the integrity of the charge air cooler coolant pump fuse. Rectify as necessary ▪ Refer to the electrical circuit diagrams and check the charge air cooler coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Check and install a new charge air cooler coolant pump only when diagnosed as failed ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Prioritized List of Possible Causes ▪ The powertrain control module has not received the expected master configuration data transmitted from the vehicle ▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Car configuration signal not received ▪ Car configuration file incorrect	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment complete network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required ▪ Clear the DTC and retest
U0402-00	Invalid Data Received From TCM - No sub type information	▪ Prioritized List of Possible Causes ▪ The powertrain control module has not received the expected CAN signal from the transmission control module within the specified time interval ▪ Implausible CAN data received from transmission control module ▪ Other transmission control module related Diagnostic Trouble Code(s) ▪ CAN harness link between powertrain control module and transmission control module network malfunction ▪ Transmission control module power and ground circuits open circuit	▪ Prioritized Checks to Perform ▪ Check transmission control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check transmission control module power and ground circuits for open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
U0402-02	Invalid Data Received From TCM - General signal failure	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0402-29	Invalid Data Received From TCM - Signal invalid	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0402-41	Invalid Data Received From TCM - General checksum failure	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0402-62	Invalid Data Received From TCM - Signal compare failure	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0402-68	Invalid Data Received From TCM - Event information	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Stop/start failure is indicated to the powertrain control module by the transmission control module ▪ Prioritized List of Possible Causes ▪ Transmission control module related concern ▪ Stop/start system related concern	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on/engine running ▪ Prioritized Checks to Perform ▪ Check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Check the stop/start system

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0402-82	Invalid Data Received From TCM - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0402-83	Invalid Data Received From TCM - Value of signal protection calculation incorrect	▪ Prioritized List of Possible Causes ▪ Transmission control module power or ground circuit open circuit, high resistance ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission system fault ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0404-41	Invalid Data Received From Gear Shift Control Module "A" - General checksum failure	▪ Prioritized List of Possible Causes ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0404-82	Invalid Data Received From Gear Shift Control Module "A" - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ Transmission control switch power or ground circuit open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Transmission control switch fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0405-68	Invalid Data Received From Cruise Control Module - Event information	▪ Prioritized List of Possible Causes ▪ The powertrain control module indicated the detection of a system event that was not caused by the powertrain control module itself but forced the powertrain control module to store the DTC, for example, missing functionality from another system or control module ▪ Other speed control system failure related Diagnostic Trouble Code(s) ▪ Other communications bus network failure related Diagnostic Trouble Code(s) ▪ Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cruise control module power or ground circuit short circuit to ground, circuit short circuit to power	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check cruise control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check communications bus network for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check speed control buttons are not jammed/contaminated/damaged. Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check cruise control module power and ground circuits for short circuit to ground, circuit short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
U0405-82	Invalid Data Received From Cruise Control Module - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ The powertrain control module has indicated that a signal was received without the corresponding rolling count value being properly updated ▪ Other speed control system failure related Diagnostic Trouble Code(s) ▪ Other communications bus network failure related Diagnostic Trouble Code(s) ▪ Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cruise control module power or ground circuit short circuit to ground, circuit short circuit to power	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check cruise control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check communications bus network for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check speed control buttons are not jammed/contaminated/damaged. Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check cruise control module power and ground circuits for short circuit to ground, circuit short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0405-84	Invalid Data Received From Cruise Control Module - Signal below allowable range	▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined failures where some circuit quantity, reported through serial data, is below a specified range ▪ Other speed control system failure related Diagnostic Trouble Code(s) ▪ Other communications bus network failure related Diagnostic Trouble Code(s) ▪ Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cruise control module power or ground circuit short circuit to ground, circuit short circuit to power	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check cruise control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check communications bus network for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check speed control buttons are not jammed/contaminated/damaged. Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check cruise control module power and ground circuits for short circuit to ground, circuit short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
U0405-86	Invalid Data Received From Cruise Control Module - Signal invalid	▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined failures where some circuit quantity, reported through serial data, is not plausible given the operating conditions ▪ Other speed control system failure related Diagnostic Trouble Code(s) ▪ Other communications bus network failure related Diagnostic Trouble Code(s) ▪ Speed control system failure connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Cruise control module power or ground circuit short circuit to ground, circuit short circuit to power	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check cruise control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check communications bus network for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Check speed control buttons are not jammed/contaminated/damaged. Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check cruise control module power and ground circuits for short circuit to ground, circuit short circuit to power ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test ▪ Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0415-02	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - General signal failure	▪ Prioritized List of Possible Causes ▪ Other anti-lock brake system control module Diagnostic Trouble Code(s) are set ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Anti-lock brake system control module power circuit failure ▪ Anti-lock brake system control module ground circuit failure	▪ Prioritized Checks to Perform ▪ Check anti-lock brake system control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0415-41	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - General checksum failure	▪ Prioritized List of Possible Causes ▪ Other anti-lock brake system control module Diagnostic Trouble Code(s) are set ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Anti-lock brake system control module power circuit failure ▪ Anti-lock brake system control module ground circuit failure	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Ignition on, engine running ▪ Prioritized Checks to Perform ▪ Check anti-lock brake system control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit to ground, short circuit to power, open circuit ▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the anti-lock brake system control module ▪ Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit
U0415-61	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Signal calculation failure	▪ Prioritized List of Possible Causes ▪ All surface progress control system fault ▪ Anti-lock brake system fault	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the terrain response switchpack for related Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0415-62	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Signal compare failure	- Prioritized List of Possible Causes - Anti-lock brake system control module power or ground circuit open circuit, high resistance - High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance - Anti-lock brake system fault	- Prioritized Checks to Perform - Refer to the electrical circuit diagrams and check the anti-lock brake system control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0415-82	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Alive/sequence counter incorrect / not updated	- Prioritized List of Possible Causes - The powertrain control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval - Implausible CAN data received from anti-lock brake system control module - Other anti-lock brake system control module related Diagnostic Trouble Code(s) - CAN harness link between powertrain control module and anti-lock brake system control module network malfunction - Anti-lock brake system control module power and ground circuits open circuit	- Vehicle Conditions to Enable DTC Logging Strategy - Ignition on, engine running - Prioritized Checks to Perform - Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest
U0415-83	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Value of signal protection calculation incorrect	- Prioritized List of Possible Causes - The powertrain control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval - Implausible CAN data received from anti-lock brake system control module - Other anti-lock brake system control module related Diagnostic Trouble Code(s) - CAN harness link between powertrain control module and anti-lock brake system control module network malfunction - Anti-lock brake system control module power and ground circuits open circuit	- Vehicle Conditions to Enable DTC Logging Strategy - Ignition on, engine running - Prioritized Checks to Perform - Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index - Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test - Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit - Using the Jaguar Land Rover approved diagnostic equipment, clear all stored Diagnostic Trouble Code(s) and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0416-46	Invalid Data Received From Vehicle Dynamics Control Module - Calibration/parameter memory failure	▪ Prioritized List of Possible Causes ▪ CAN network failure ▪ Invalid CAN data received from anti-lock brake system control module	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module ▪ Clear the DTC and retest
U0416-68	Invalid Data Received From Vehicle Dynamics Control Module - Event information	▪ Prioritized List of Possible Causes ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module ▪ Clear the DTC and retest	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair harness as required ▪ Check powertrain control module for additional Diagnostic Trouble Code(s) and refer to relevant DTC index. Rectify these first ▪ Check anti-lock brake system control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the anti-lock brake system control module ▪ Clear the DTC and retest
U0417-41	Invalid Data Received From Park Brake Control Module - General checksum failure	▪ Prioritized List of Possible Causes ▪ Electric park brake control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the electric park brake control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the electric park brake control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0417-82	Invalid Data Received From Park Brake Control Module - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ Electric park brake control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Electric park brake system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the electric park brake control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the electric park brake control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0426-00	Invalid Data Received From Vehicle Immobilizer Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ Harness fault - CAN circuit ▪ Incorrect target ID written to powertrain control module ▪ Body control module failure	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a network integrity test ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required ▪ Using the Jaguar Land Rover approved diagnostic equipment complete immobilization application ▪ Check the body control module control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0426-31	Invalid Data Received From Vehicle Immobilizer Control Module - No signal	▪ Prioritized List of Possible Causes ▪ Harness fault - CAN circuit ▪ Incorrect target ID written to powertrain control module ▪ Body control module failure	▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a network integrity test ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required ▪ Using the Jaguar Land Rover approved diagnostic equipment complete immobilization application ▪ Check the body control module control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U042B-29	Invalid Data Received From Chassis Control Module "A" - Signal invalid	▪ Prioritized List of Possible Causes ▪ Suspension control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Suspension system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the suspension control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment check the suspension control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U042B-68	Invalid Data Received From Chassis Control Module "A" - Event information	▪ Prioritized List of Possible Causes ▪ Suspension control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Deployable spoiler system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the suspension control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment check the suspension control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U042B-82	Invalid Data Received From Chassis Control Module "A" - Alive/sequence counter incorrect / not updated	▪ Prioritized List of Possible Causes ▪ Suspension control module power or ground circuit open circuit, high resistance ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Suspension system fault	▪ Prioritized Checks to Perform ▪ Refer to the electrical circuit diagrams and check the suspension control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary ▪ Using the Jaguar Land Rover approved diagnostic equipment check the suspension control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
U0447-00	Invalid Data Received From Gateway "A" - No sub type information	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity dual battery system ▪ Harness failure - Wiring integrity gateway module	▪ Prioritized Checks to Perform ▪ Check gateway module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Clear the DTC and retest
U0452-00	Invalid Data Received From Restraints Control Module - No sub type information	▪ Prioritized List of Possible Causes ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Harness failure - Wiring integrity seatbelt sensor ▪ Harness failure - Wiring restraints control module	▪ Prioritized Checks to Perform ▪ Check restraints control module for related Diagnostic Trouble Code(s) and refer to relevant DTC index ▪ Inspect connectors for signs of water ingress and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Clear the DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U0452-64	Invalid Data Received From Restraints Control Module - Signal plausibility failure	▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ Yes
		▪ Control Module Cavity
		  ▪ Circuit reference - I_S_INS -
		▪ Prioritized List of Possible Causes
		  ▪ Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Restraints control module power or ground circuit open circuit, high resistance
		  ▪ Supplementary restraint system fault
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Refer to the electrical circuit diagrams and check the restraints control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

U0592-00	Invalid Data Received From Gear Shift Control Module "B" - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Transmission control switch power or ground circuit open circuit, high resistance
		  ▪ High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Transmission control switch fault
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the transmission control switch power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the transmission control switch for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

U0601-08	Lost Communication With Particulate Filter Pressure Sensor "A" - Bus signal/message failures	▪ MIL
		  ▪ Yes
		▪ Electrical Cause
		  ▪ Yes
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion
		▪ Gasoline particulate filter pressure sensor - Wiring integrity
		  ▪ Fuse failure
		  ▪ Short circuit to ground
		  ▪ Short circuit to power
		  ▪ Open circuit
		  ▪ High resistance
		▪ Powertrain control module failure
			▪ Vehicle Conditions to enable DTC Logging Strategy
			  ▪ Ignition on
			▪ Prioritized Checks to Perform
			▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
			▪ Refer to the electrical circuit diagrams and check the gasoline particulate filter pressure sensor circuit for
			  ▪ Fuse failure
			  ▪ Short circuit to ground
			  ▪ Short circuit to power
			  ▪ Open circuit
			  ▪ High resistance
			▪ Check and install a new powertrain control module, only when diagnosed as failed

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U1A00-00	Private Communication Network - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Harness fault - LIN circuit
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check LIN circuits. Repair wiring harness as required

U1A14-00	CAN Initialization Failure - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Harness fault - CAN circuit
		  ▪ Powertrain control module - Connector is disconnected, connector pin is backed out, connector pin corrosion
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment complete a network integrity test
			  ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required. Inspect connectors for signs of water ingress and pins for damage and/or corrosion

U2006-49	Network Controller - Internal electronic failure	▪ Prioritized List of Possible Causes
		  ▪ PCM power failure
		  ▪ FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the power and ground connections to the powertrain control module
			  ▪ Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Replace the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the adaptive cruise control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

U2012-00	Car Configuration Parameter(s) - No sub type information	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Monitor Description
		  ▪ Any car configuration parameter received from body control module is outside the designated limits allowed
		▪ Prioritized List of Possible Causes
		  ▪ Body control module not transmitting some or all of the car configuration CAN data
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Harness fault - CAN circuit
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required
			  ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required
			  ▪ Check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Clear the DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U2012-02	Car Configuration Parameter(s) - General signal failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

U2012-29	Car Configuration Parameter(s) - Signal invalid	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

U2012-64	Car Configuration Parameter(s) - Signal plausibility failure	▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module detected plausibility failures
		  ▪ Car configuration signal not received
		  ▪ Car configuration file incorrect
		  ▪ Harness fault - CAN circuit
		  ▪ Body control module not transmitting some or all of the car configuration CAN data
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment check and up-date the car configuration file as required
			  ▪ Refer to the electrical circuit diagrams and check CAN circuits. Repair wiring harness as required
			  ▪ Check the body control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Clear the DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U2012-86	Car Configuration Parameter(s) - Signal invalid	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

U2108-00	Adaptive Cruise Control - No sub type information	▪ Prioritized List of Possible Causes
		  ▪ Adaptive speed control system failure - Error indicating adaptive speed control failure flag set
		  ▪ Adaptive cruise control module power or ground circuit open circuit, high resistance
		  ▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance
		  ▪ Adaptive speed control system fault
			▪ Prioritized Checks to Perform
			  ▪ Refer to the electrical circuit diagrams and check the adaptive cruise control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, check the adaptive cruise control module for related Diagnostic Trouble Code(s) and refer to the relevant DTC index

U2108-24	Adaptive Cruise Control - Signal stuck high	▪ Prioritized List of Possible Causes
		  ▪ The powertrain control module measures a signal that remains high when changes are expected
		  ▪ Adaptive speed control system failure - Adaptive speed control follow speed error
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required
			▪ Prioritized Checks to Perform
			  ▪ Check adaptive cruise control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required
			  ▪ Clear the DTC and retest

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U2108-64	Adaptive Cruise Control - Signal plausibility failure	▪ Prioritized List of Possible Causes ▪ The powertrain control module detected plausibility failures ▪ Adaptive speed control system failure - Adaptive speed control follow speed range error	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check adaptive cruise control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required ▪ Clear the DTC and retest
U2108-68	Adaptive Cruise Control - Event information	▪ Prioritized List of Possible Causes ▪ Adaptive speed control system failure - Error indicating adaptive speed control follow speed check when stationary	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check adaptive cruise control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required ▪ Clear the DTC and retest
U2108-86	Adaptive Cruise Control - Signal invalid	▪ Prioritized List of Possible Causes ▪ The powertrain control module has determined failures where some circuit quantity, reported through serial data, is not plausible given the operating conditions ▪ Adaptive speed control system failure - Error when invalid adaptive speed control resume requests are present	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Turn Ignition on and wait for a few seconds. Active cruise control is not required ▪ Prioritized Checks to Perform ▪ Check adaptive cruise control module for Diagnostic Trouble Code(s) and refer to the relevant DTC index ▪ Using the Jaguar Land Rover approved diagnostic equipment, complete a CAN network integrity test. Refer to electrical circuit diagrams and check CAN circuits if required. Repair wiring harness as required ▪ Clear the DTC and retest
U2300-51	Central Configuration - Not programed	▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Prioritized List of Possible Causes ▪ The car configuration file has never been received from the master control module by a newly installed powertrain control module ▪ Incorrect calibration installed into the powertrain control module	▪ Vehicle Conditions to Enable DTC Logging Strategy ▪ Clear the DTC ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes) ▪ Cycle Ignition ▪ Read Diagnostic Trouble Code(s) ▪ Prioritized Checks to Perform ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software ▪ Check calibration part number is correct for the vehicle

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION

U2300-54	Central Configuration - Missing calibration	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

U2300-56	Central Configuration - Invalid/incomplete configuration	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ The car configuration file data being transmitted to the powertrain control module is incorrect and contains zero value parameters where a non zero value is required
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

U2300-64	Central Configuration - Signal plausibility failure	▪ Electrical Cause
		  ▪ No
		▪ Mechanical Cause
		  ▪ No
		▪ Prioritized List of Possible Causes
		  ▪ Incorrect car configuration file installed into the powertrain control module
		  ▪ Incorrect calibration installed into the powertrain control module
			▪ Vehicle Conditions to Enable DTC Logging Strategy
			  ▪ Clear the DTC
			  ▪ Allow sufficient time for master control module to begin transmitting car configuration file data and modules to accept and receive data (5 minutes)
			  ▪ Cycle Ignition
			  ▪ Read Diagnostic Trouble Code(s)
			▪ Prioritized Checks to Perform
			  ▪ Using the Jaguar Land Rover approved diagnostic equipment, reconfigure the master control module with the latest level software
			  ▪ Check calibration part number is correct for the vehicle

cmVlX2FtZzsyMDI1LTEyLTE5VDA3OjAzOjI2LjM4MIo7MjA3LjE4MS4zLjEzMTtTQUpERDFHWDRMQ0s2NzQ5Nw==